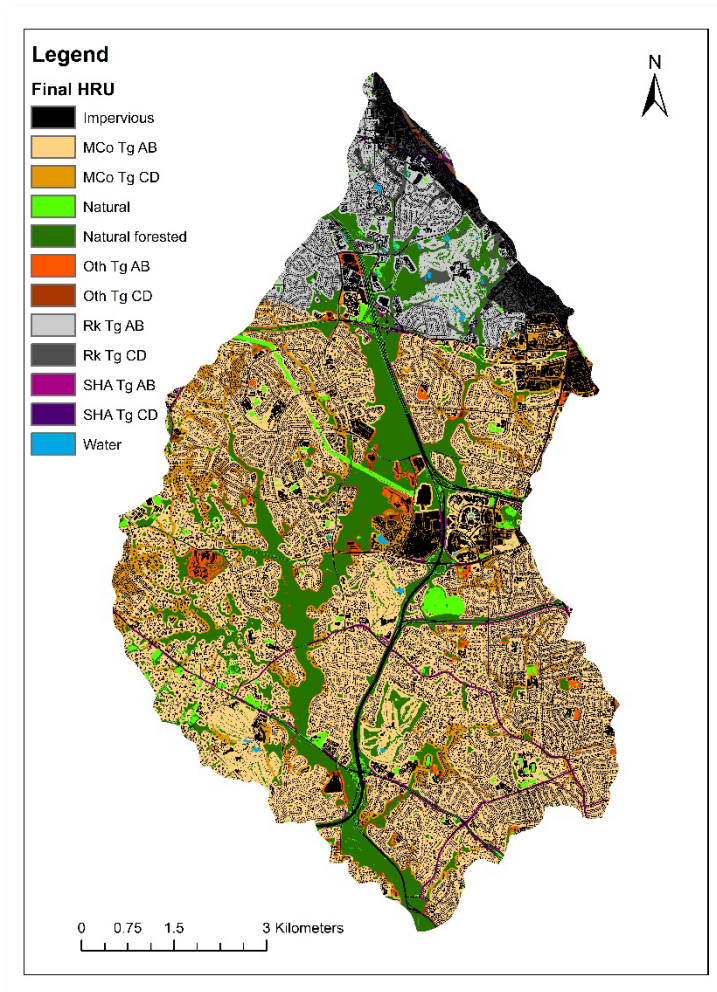




WMOST v3 Case Study: Cabin John Creek, Maryland



Cabin John Creek Hydrologic Response Units

WMOST v3 Case Study: Cabin John Creek, Maryland

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Notice and Disclaimer

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Abstract

A case study application of EPA's Watershed Management Optimization Support Tool (WMOST v3) was carried out to inform management options for the heavily urbanized watershed of Cabin John Creek (CJC), MD, a tributary to the Potomac River, for the 10-year period 2014 - 2025. CJC was chosen as a representative case study of a watershed required to meet loading targets for both the Chesapeake Bay Total Maximum Daily Load (TMDL) as well as a Maryland nontidal stream TMDL for total suspended solids (TSS), but without water supply constraints. WMOST is an application designed to facilitate cost-effective integrated water resource management at the scale of 12- to 10-digit Hydrologic Units (HUC12 – HUC10; 10,000 – 250,000 acres). Optimizations were performed to meet the single least-cost objective, while meeting one of the constraints imposed by total nitrogen and total phosphorus load reduction targets (4% TN, 5% TP) associated with the Chesapeake Bay TMDL or the TSS loading reduction targets (21%) for a CJC non-tidal sediment TMDL, implemented either as a TSS load constraint or as an associated flow target. Alternative approaches to deriving TSS loading targets were explored: reduction of 1) maximum modelled daily load for a given year, 2) confidence limits for maximum modelled daily load based on historic temporal variability, 3) peak flow targets based on simulation of pre-development conditions (1-year 24-hour event), and 4) peak flow targets associated with predicted TSS load based on sediment rating curves. Decision variables included the type and implementation level of upland management (seven structural stormwater control measures [SCMs], nonstructural urban tree canopy planting, riparian buffer restoration), instream measures (outfall enhancement, stream restoration), and programmatic measures (street sweeping). Baseline unit runoff, recharge, and loading time series were derived from the Beta 3 version of the Phase 6 Chesapeake Bay Watershed Model, adjusted to distribute loads between two hydrologic soil groups (A + B, C + D) for all developed hydrologic response units. Costs included annualized capital costs for design and construction of SCMs or other BMPs (based on actual cost data from Montgomery County, MD, the city of Rockville, and the MD State Highway Administration) plus operation and maintenance costs based on default values from EPA's System for Urban Stormwater Treatment and Analysis Integration (SUSTAIN) tool. Robustness of solutions was tested by comparing results for a year with above-average annual precipitation (2003) with results for 2014, a year with 100-year average annual precipitation but one large event. The flow target that was set based on estimates of pre-development 1-year 24-hour events (11 - 13 cfs), was not achievable with any level of SCM or BMP implementation. Peak flow targets based on application of sediment rating curves (499 cfs in 2014, 535 cfs in 2003) were also not achievable. The minimum achievable targets based on incremental scenarios of steadily decreasing peak flow targets were 700 cfs for the 2014 weather regime and 525 cfs for the 2003 weather regime. These flow targets required implementation of a combination of sand filters, infiltration basins, and porous pavement for both 2003 and 2014 weather

regimes. Ancillary benefits of this approach included significant increases in predicted baseflows. The flow-optimized solutions would increase baseflow the greatest amount during the growing season (456 - 517%), and to a lesser extent during fall and winter months. The least-cost solutions chosen to achieve a 21% reduction in annual and maximum daily TSS loads included implementation of infiltration basins, dry pond to wet pond conversions, and sand filters for both 2003 and 2014 weather regimes. Additional SCMs and level of implementation were required to achieve maximum daily TSS loads adjusted for a 95% confidence interval – a combination of infiltration basins, dry pond to wet pond conversions, sand filters, bioretention basins, and riparian buffer restoration. Solutions based on the maximum daily load with a 95% confidence interval yielded similar but slightly higher peak flows than minimum achievable levels. Implementation of least-cost solutions to meet TSS loading targets would yield load reductions for total N (17.4 – 17.9%) and total P (16.6 – 19.9%) that far exceed load targets for those nutrients (5% TN, 4% TP). Meeting sediment load reduction targets will require greater implementation of green infrastructure stormwater control measures, particularly those with enhanced infiltration. Modelling results show limited effectiveness for gray infrastructure (extended detention basins) in meeting water quality goals.

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We would like to acknowledge the assistance of Montgomery County, the city of Rockville, and MD State Highway Administration in compiling actual costs associated with design and construction of stormwater control measures and other best management practices in the study region.

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Acronyms

A+B – Soil hydrologic groups A plus B
AMPL – algebraic model programming language
ASR – aquifer storage and recovery
AWQMS – Ambient Water Quality Monitoring System
BMP – best management practice
C+D – Soil hydrologic groups C plus D
CB – Chesapeake Bay
CBWM – Chesapeake Bay Watershed Model
Cfs – cubic feet per second
CJC – Cabin John Creek
CN – Curve Number
CV – Coefficient of variation
DEM – digital elevation model
DQgsw – WMOST decision variable for flow from groundwater to surface water
EIA – effective impervious area
EPA – Environmental Protection Agency
GIS – geographic information system
GW – groundwater
HRU – hydrologic response unit
HUC – hydrologic unit
HUC10 – 10-digit hydrologic unit
HUC12 – 12-digit hydrologic unit
Inf – infeasible result
INFIL – infiltration rate
Lb – pound
LTA – Long term average annual load
MCo – Montgomery County
MD – Maryland
MDE – Maryland Department of Environment
MDL – maximum daily load
MG – million gallons
MGD – million gallons per day
MS4 – municipal separate storm sewer systems
Mt – metric ton
Mtg – turfgrass HRU
NEOS – Network-Enabled Optimization System
NHD – National Hydrography Dataset
NSE – Nash Sutcliffe Efficiency coefficient
O&M – operations and maintenance cost

ORISE – Oak Ridge Institute for Science Education
Oth – Other regulated parties
P – precipitation
PET – potential evapotranspiration
RI – Rhode Island
RO – runoff
Rv – city of Rockville
SCM – stormwater control measure
SHA – State Highway Administration
STST – streambank stabilization
STSW – street sweeping
SUSTAIN - System for Urban Stormwater Treatment and Analysis Integration
SW – southwest
SWMM – stormwater management model
TC – tree canopy
Tg – turfgrass
TMDL – total maximum daily load
TN – total nitrogen
TP – total phosphorus
TR-55 – Technical Release 55 Urban Hydrology for Small Watersheds Model
TSS – total suspended solids
UAW – unaccounted water
UD – underdrain
US EPA – United States Environmental Protection Agency
USGS – United States Geological Survey
WIP – Watershed implementation plan
WMOST – Watershed Management Optimization Support Tool
WWTP – wastewater treatment plant
Z – z score

Introduction

Problem Definition

Governments at all levels are faced with the challenge of finding cost-effective solutions to meet water quality objectives, whether to meet water quality criteria or to meet loading targets such as Total Maximum Daily Loads (TMDLs) established under Section 303(d) of the Clean Water Act to bring impaired water bodies back into compliance. The U.S. Environmental Protection Agency (US EPA) established the Chesapeake Bay (CB) TMDL on December 29, 2010, to restore conditions in the Bay as well as upstream waters (US EPA 2010). The TMDL identifies the necessary pollution reductions from major sources of total nitrogen (TN), total phosphorus (TP) and total suspended sediment (TSS) across the Bay jurisdictions and sets pollution limits necessary to meet water quality standards. Bay jurisdictions include Delaware, Maryland, New York, Pennsylvania, Virginia, West Virginia and the District of Columbia. The TMDL is designed to ensure that all pollution control measures needed to fully restore the Bay and its tidal rivers are in place by 2025. The TMDL also calls for practices to be in place by 2017 to meet 60 percent of the overall nitrogen, phosphorus and sediment reductions.

Many of the planned reductions in TN and TP in the TMDL Phase 1 Watershed Implementation Plans (WIPs) were achieved through point source controls on discharges from wastewater treatment plants (WWTP) and, to a lesser extent, through agricultural best management practices (BMPs). Planned reductions in the Phase II and Phase III WIPs rely much more heavily on reduction of urban runoff through stormwater control measures. The runoff and septic leaching from urban and suburban land comprise about 20% of the total nitrogen load in Maryland (CSN 2011). To achieve the target load, more than 2.5 million pounds of nitrogen need to be reduced from the urban sector, equivalent to a 37% reduction of nitrogen coming from existing development in the state. Stormwater runoff from urban and suburban land also comprises about 20% of the total phosphorus load in Maryland. To achieve the target load for phosphorus, more than a quarter million pounds will need to be reduced, equivalent to a 36% reduction in phosphorus load from existing development in the state (CSN 2011).

The state of Maryland is responsible for meeting not only targets for the Chesapeake Bay TMDL, which were set to achieve applicable water quality standards in the Chesapeake Bay, its tidal rivers and embayments, but also TMDLs for impaired non-tidal waters in the state. For example, the Maryland Department of Environment (MDE) has identified the waters of the Cabin John Creek watershed within the State's 2010 Integrated Report as impaired by sediments (1996), phosphorus (1996), bacteria (2002), chlorides (2010), sulfates (2010), habitat alteration, and impacts to biological communities (2006) (MDE 2010).

Phosphorus is now listed as meeting criteria. The sediment TMDL established for Cabin John Creek (CJC), based on a total baseline sediment load of 5,537.7 tons/year, is 4,391.4 tons/year of sediment (total suspended solids) (MDE 2011). Meeting the CJC sediment TMDL target requires implementation of stormwater control measures by municipal (Rockville), county (Montgomery County, MD) and state (MD State Highway Administration) agencies, among others.

Watershed Management Optimization Tool

The U.S. EPA has created the Watershed Management Optimization Support Tool (WMOST) to facilitate cost-effective integrated water resources management at the local community and watershed scales, typically a 12- or 10-digit Hydrologic Unit Code (HUC; 10,000 – 250,000 acres; USGS and USDA NRCS 2013). WMOST version 3.01 allows users to find the least-cost alternative among a suite of potential management practices spanning stormwater, wastewater, drinking water, and land conservation options, to meet both water quantity and water quality goals (Detenbeck et al. 2018a,b,c, Detenbeck and Weaver 2019). During development of WMOST v3, US EPA Office of Research and Development sought partners to help in design, evaluation and testing to ensure that the final tool would meet users' needs. MDE was identified as a partner applying WMOST v3.01 within Cabin John Creek, Montgomery County, MD, as a case study for testing and development.

WMOST uses baseline time series for unit area runoff, recharge and pollutant loadings based on outputs from an externally-run watershed model. The effect of stormwater control measures is simulated through connections to the nonGIS version of EPA's System for Urban Stormwater Treatment and Analysis Integration (SUSTAIN; US EPA 2009) tool which uses algorithms from EPA's Storm Water Management Model (SWMM; Rossman 2015) to estimate reductions in unit area runoff and pollutant loads and corresponding increases in infiltration. WMOST defines the optimization problem using the algebraic model programming language (AMPL). In formulating the optimization problem, the objective is defined as the (minimization of) the sum of annualized capital costs and annual operations and maintenance costs over the planning period while meeting mass-balance constraints as well as user-defined constraints such as daily or annual loading targets. (See WMOST v3 Theoretical Documentation for more detail.) (Detenbeck et al. 2018c) The optimization problem is then submitted to the online Network-Enabled Optimization System (NEOS) server (<https://neos-guide.org/>) for a solution using a mixed integer nonlinear programming method (Bonami et al. 2008). Results files, including least-cost management options, are imported back into WMOST and displayed for the user in graphical and tabular form (Figure 1).

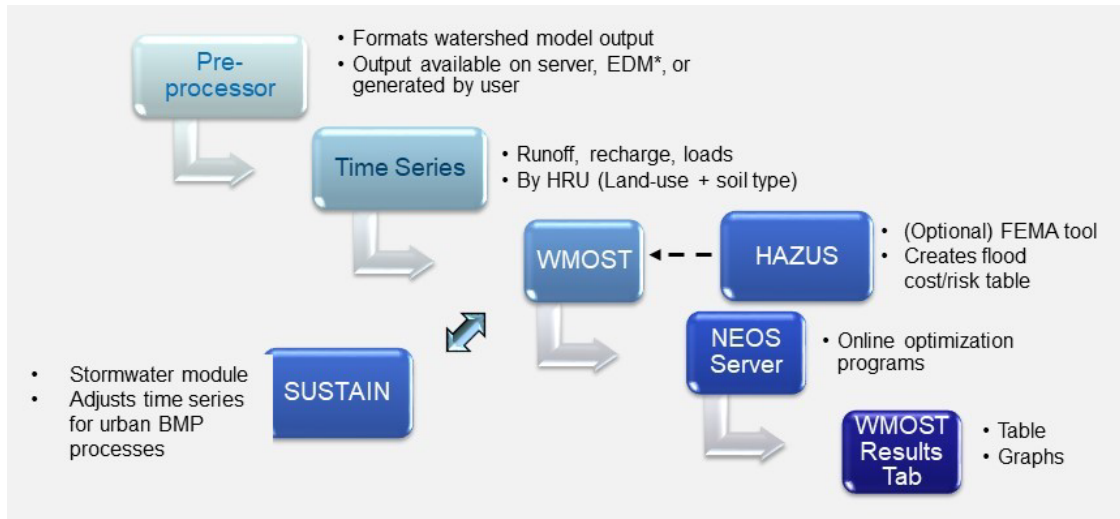


Figure 1. Flow of data and information into and out of WMOST. *EDM = EPA Estuary Data Mapper.

Objectives

The main objective of this case study is to apply WMOST v3 to find the most cost-effective suite of best management practices to meet both the Chesapeake Bay TN, TP, and TSS loading targets for the Bay TMDL as well as non-tidal sediment load targets based on the CJC TMDL. Sub-objectives include 1) an evaluation of the robustness of optimal management strategies under varying weather regimes, 2) a comparison of optimal management practices under different approaches for establishing sediment TMDL targets, and 3) an evaluation of the implications of applying optimal management approaches for suspended sediment goals for meeting the Bay TMDL goals for TN and TP (Table 1). This study is designed as the first in a series of MD case studies with WMOST for different use cases, to inform not only cost-effective management of a particular watershed, but also to provide guidance for communities and agencies dealing with similar goals and environmental settings. This initial use case represents a watershed needing to meet both non-tidal and downstream loading targets through stormwater controls, but without the added complications of significant point sources or water quantity constraints. Water quantity is not as constrained within the CJC watershed because communities in the watershed meet their drinking water supply needs and wastewater treatment needs through interbasin transfers.

Table 1. Summary of WMOST runs needed to accomplish multiple sub-objectives related to Chesapeake Bay and Cabin John Creek TMDLs

WMOST Run	Year	Type of Setup	Parameter	SubObjective
1a	2006	Calibration	TSS	Calibrate WMOST for year with average precipitation
1b	2006	Calibration	TP	Calibrate WMOST for year with average precipitation
1c	2006	Calibration	TN	Calibrate WMOST for year with average precipitation
2a	2014	Validation	TSS	Establish baseline for optimizations
2b	2014	Validation	TP	Establish baseline for optimizations
2c	2014	Validation	TN	Establish baseline for optimizations
3a	2014	Optimization	TP	Evaluate optimal management strategies to meet Bay TMDL maximum daily loading target for TP, including potential stormwater control measures and riparian buffer management for baseline year with average precipitation
3b	2014	Optimization	TN	Evaluate optimal management strategies to meet Bay TMDL maximum daily loading target for TN, including potential stormwater control measures and riparian buffer management for baseline year with average precipitation
3c	2014	Optimization	TSS	Evaluate optimal management strategies to meet Bay TMDL maximum daily loading target for TSS, including potential stormwater control measures and riparian buffer management for baseline year with average precipitation
3d	2014	Optimization	TSS	Evaluate optimal management strategies to meet Bay TMDL maximum daily loading target for TSS, including alternative nonstructural BMPs for baseline year with average precipitation
4a	2003	Optimization	TP	Evaluate robustness of optimal management strategies to meet Bay TMDL maximum daily loading target for TP, including potential stormwater control measures and riparian buffer management for relatively wet year to compare with baseline year (average precipitation)

Table 1. (continued)

WMOST Run	Year	Type of Setup	Parameter	SubObjective
4b	2003	Optimization	TN	Evaluate robustness of optimal management strategies to meet Bay TMDL maximum daily loading target for TN, including potential stormwater control measures and riparian buffer management for relatively wet year to compare with baseline year (average precipitation)
4c	2003	Optimization	TSS	Evaluate robustness of optimal management. strategies to meet Bay TMDL maximum daily loading target for TSS, including potential stormwater control measures for relatively wet year to compare with baseline year (average precipitation)
4d	2003	Optimization	TSS	Evaluate robustness of optimal management strategies to meet Bay TMDL maximum daily loading target for TSS, including potential riparian management and streambank stabilization measures for relatively wet year to compare with baseline year (average precipitation)
5a	2014	Optimization	TSS (via flow target)	Compare optimal management. strategies to meet non-tidal CJC TMDL via flow-based target with optimal management strategy via load-based target for baseline year (average precipitation)
5b	2003	Optimization	TSS (via flow target)	Evaluate robustness of optimal management. strategies to meet Bay TMDL maximum daily loading target for TSS (flow-based), including potential riparian management and streambank stabilization measures for relatively wet year to compare with baseline year (average precipitation)
6a	2014	Simulation	TP	Evaluate whether optimal management. strategies to meet non-tidal sediment TMDL target for CJC would meet Bay TMDL maximum daily loading target for TP for baseline year with average precipitation
6b	2014	Simulation	TN	Evaluate whether optimal management. strategies to meet non-tidal sediment TMDL target for CJC would meet Bay TMDL maximum daily loading target for TN for baseline year with average precipitation

Study Area Characteristics

Cabin John Creek, located in Montgomery County, MD, is a tributary to the Potomac River, which, in turn, is one of the main tributaries to the Chesapeake Bay (Figure 2). The CJC watershed covers approximately 67 square kilometers, comprising a single 12-digit Hydrologic Unit (HUC), with a population of approximately 75,170 (MDE 2011). The watershed is contained within the Piedmont geologic province, with gentle to steep rolling topography. The watershed is comprised of mainly hydrologic soil group B and C type soils with lesser coverage of type A and D soils (Figure 3a). Based on the CB Watershed Model 5.2 Land Use categories, the CJC watershed is covered mainly by urban land uses with less than 10% forest cover, and minimal amounts of agriculture (0.6%; MDE 2011; Figure 3b). Some stormwater control measures already have been implemented in the watershed, primarily wet ponds (treating 432.6 ha [1069 acres] or 7.1% of developed land in the watershed) and extended dry detention basins (treating 401 ha [991 acres] or 6.6% of developed land) (Jeff White, MDE, pers. comm.).

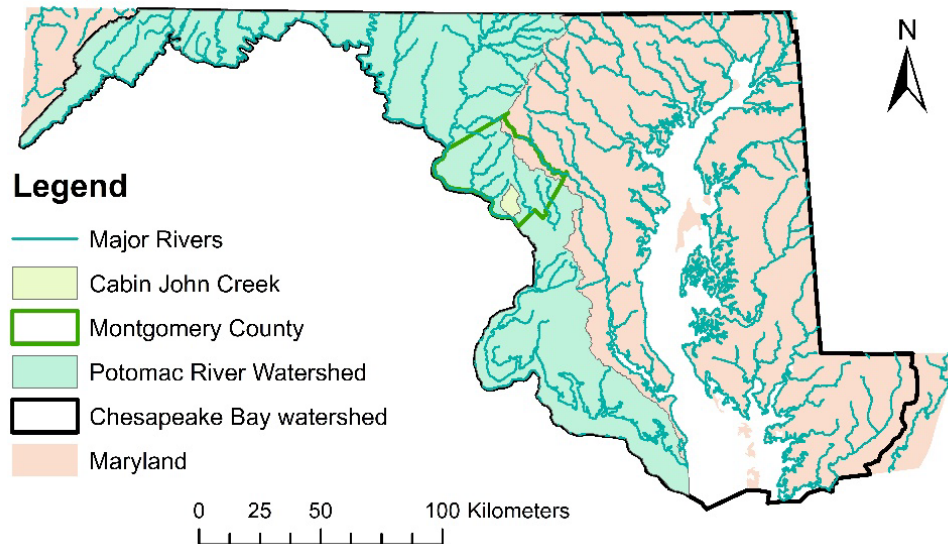


Figure 2. Location of Cabin John Creek watershed within Potomac River watershed, Chesapeake Bay watershed, and Montgomery County, MD boundaries. The Potomac River watershed is shown in green, with the Cabin John Creek subwatershed shown in light green.

Methods

Data sources required for setting up WMOST model runs are summarized in Appendix B.

Optimization Problem: Objective Function, Constraints, Decision Variables

WMOST v3 was used to define the optimization problem for the CJC case study. WMOST defines a single-objective optimization problem, with cost minimization over the planning period as the goal. Costs include both initial capital costs, annualized over the 10-year planning period (2015 – 2025) based on a given interest rate (5% in our example), as well as annual operations and maintenance costs. Multiple mass-balance constraints are defined within WMOST to describe hydrologic flows and pollutant loadings (Detenbeck et al. 2018b). In addition, user-defined constraints can be set to represent water quality and quantity targets. In this case, the Bay TMDL TN, TP, and TSS targets (5%, 4%, and 4% reductions) and non-tidal CJC sediment TMDL targets (21% reductions in TSS load) were used to define loading constraints. Percent reductions were applied to WMOST-calibrated 2015 TN, TP and TSS loads for CJC to determine target loads (Method 1). Outputs from the CBWM for 2015 for the Montgomery County land-water segment were used as the basis for WMOST 2015 calibrations, with some adjustments made to better match local hydrology measurements. For comparison, an alternative approach for establishing sediment targets calculated a peak flow target corresponding to a sediment daily loading target generated using MDE methods (based on a correction factor for daily loads designed to account for temporal variability) and a sediment rating curve established for the nearby Anacostia River (see below for details; Method 2a-c).

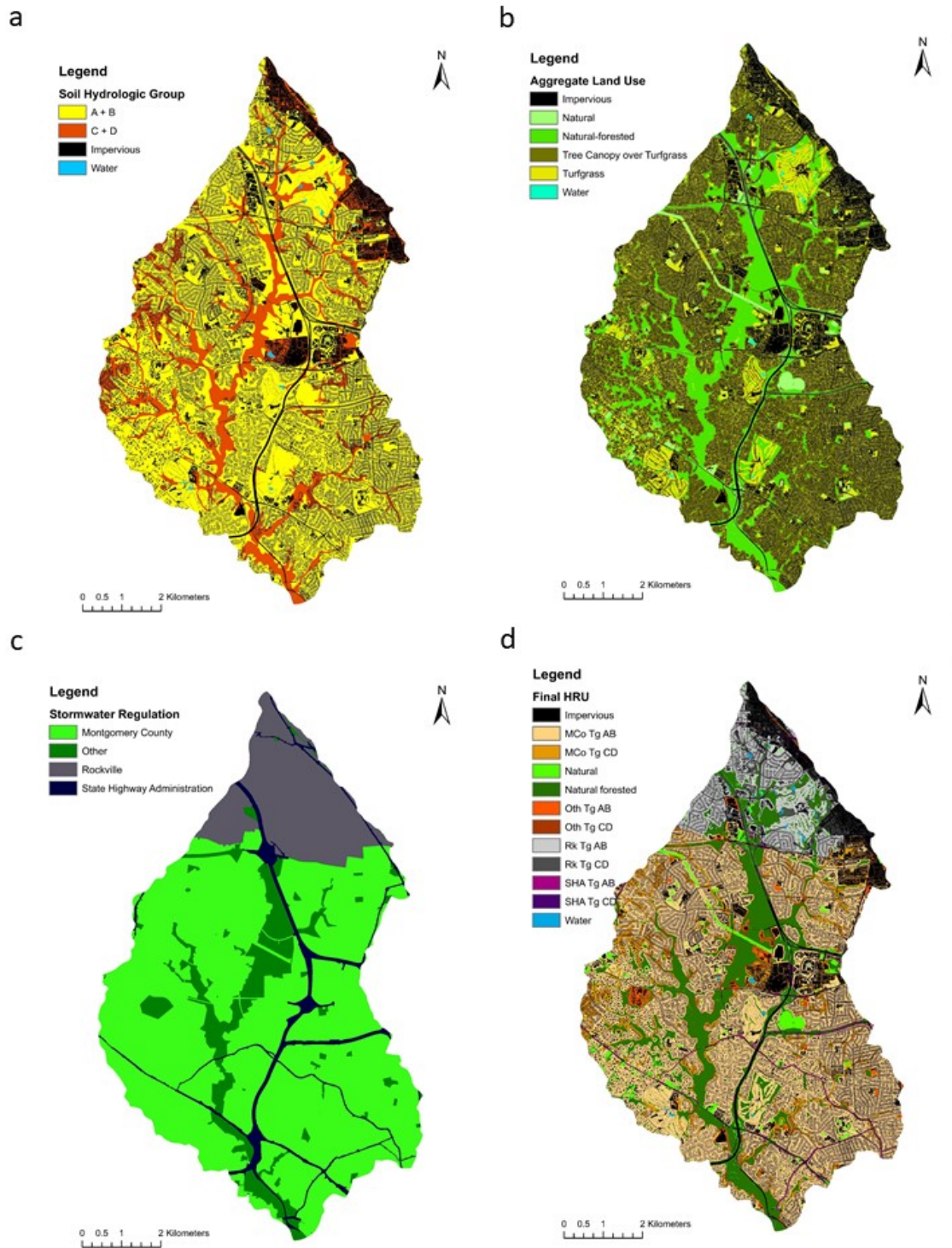


Figure 3. a) Soil Hydrologic Groups, b) Aggregate Land Use Classes, c) Stormwater Regulation Entity, and d) Final Hydrologic Response Units (HRUs). MCo = Montgomery County, Rv = Rockville, SHA = State Highway Administration, Oth = Other, AB = Type A + B hydrologic soil groups, CD = C + D hydrologic soil groups, Tg = Turfgrass. See Appendices 1a-d for full size maps.

Decision variables allowed to vary during optimization runs included areas of developed land subjected to treatment by individual BMPs, riparian buffer restoration area, implementation of alternative BMPs (stream bank restoration, outfall enhancements), or nonstructural BMPs (nutrient management plan, street sweeping, tree canopy over turf).

Conceptual Model for System

Figure 4a illustrates the WMOST conceptual model describing the interaction of natural features in a watershed with manmade infrastructure when determining water and pollutant flows. Figure 4b illustrates the simplified system for CJC, with various components not present in the CJC watershed WMOST model removed. Most of the infrastructure components could be omitted from the WMOST model because drinking water sources and wastewater treatment were located outside of the watershed and thus, did not affect the water or pollutant load balances.

HRU Definitions

WMOST is a semi-lumped parameter model in which runoff, recharge, and pollutant loads are tracked for aggregate land areas within a watershed. These lumped land areas are known as Hydrologic Response Units (HRUs). In watershed models, HRUs are typically defined by combinations of land-use, soils, and sometimes slope classes as they are areas with similar runoff/recharge/loading characteristics. Appendix C lists the original land-use classes for the Montgomery County Land/Water segment defined to support the Chesapeake Bay Watershed Model (CBWM) v6 (<http://cast.chesapeakebay.net/>). The WMOST application is more likely to reach an optimal solution within convergence time limits established by the NEOS optimization server with fewer HRUs (e.g., 10 – 12), so we first collapsed classes from the original CBWM set into Impervious, Turfgrass, Natural, and Water categories (Table 2, Figure 3b). Second, we disaggregated the developed land classes based on permit areas. The permit areas are based on a regulated stormwater shapefile developed by MDE in conjunction with their MS4 program and the regulated jurisdictions/entities (Figure 3c: City of Rockville, Montgomery County, MD State Highway Administration, Other regulated) to allow consideration of different site constraints and BMP costs by land owner/Municipal Separate Storm Sewer (MS4) permit. Third, because WMOST is designed to work with either pervious or blended pervious + impervious HRUs, we created combined pervious + impervious classes for the developed HRUs. Finally, the CBWM does not distinguish between soil hydrologic groups in defining HRUs. However, soil hydrologic groups are important in defining siting constraints for stormwater control measures in Maryland (CWP and MDE 2009), and soil infiltration rates strongly influence the effectiveness of SCMs in reducing runoff and pollutant loads (TetraTech 2010). Therefore, we further disaggregated HRUs by soil hydrologic group from the Soil Survey Geographic Database (SSURGO) soil layers (USDA NRCS 2013): A/B vs C/D to create the final set of 11 HRU classes (Table 2, Figure 3d). In some analyses, natural forested and nonforested HRUs were combined, but when assessing

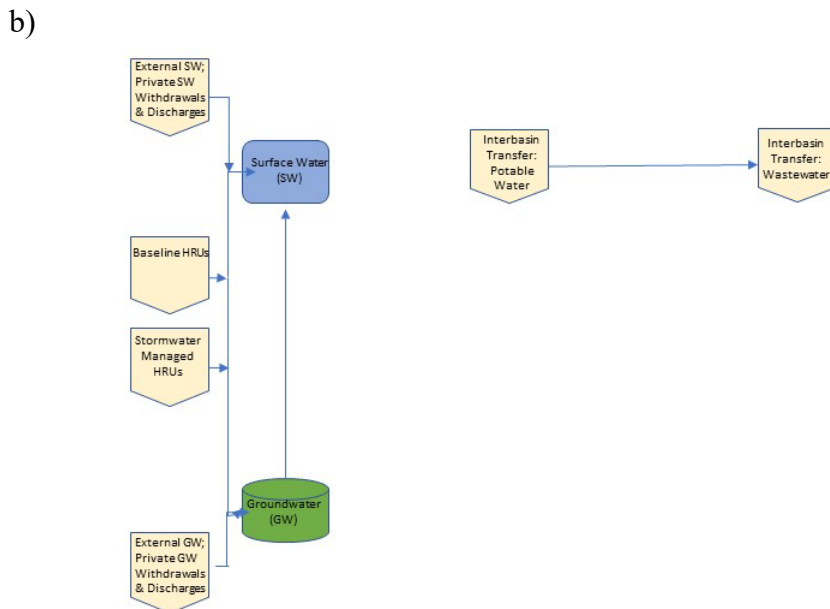
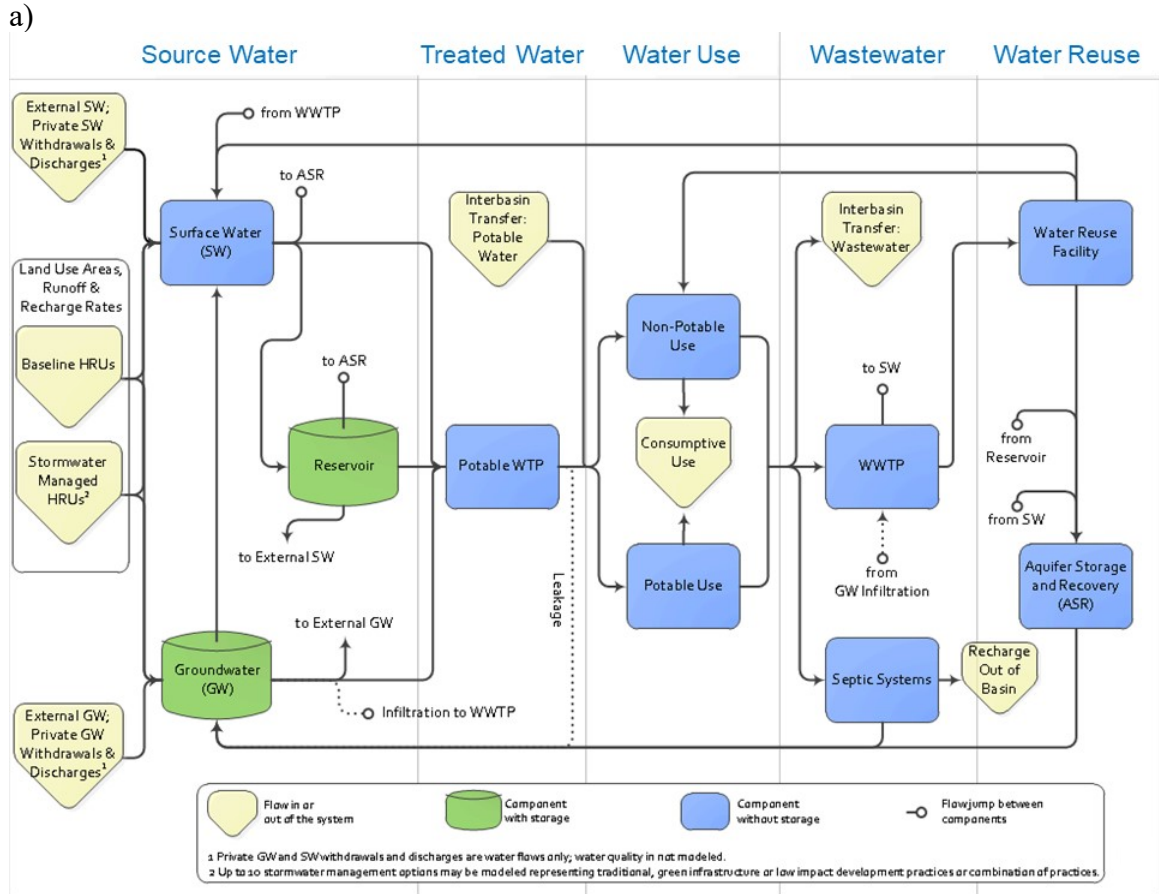


Figure 4. a) Water and pollutant sources and components and flows among components in WMOST, including both natural and manmade features. b) Simplified components and flow routing in WMOST for Cabin John Creek watershed.

riparian buffer restoration, these were kept separate. Likewise, tree canopy over turfgrass was kept separate from turf grass when considering tree canopy as an alternative BMP.

Baseline Time Series

We derived baseline time series of weather, hydrology, and pollutant loads from datasets used or generated by the hybrid CBWM v6 (beta 3). Time series were provided by the Chesapeake Bay Program Office (Ghopal Bhatt, pers. comm.). (The final version 6 model is now available, but the beta 3 model was the most recent version available at the time this project was initiated.) We used the forested land-use class time series (for) and non-forested category (osp) time series, combined in proportion to their respective areas, to represent the Natural HRU. For the pervious component of developed lands, we used the time series for turfgrass (mtg). We calculated expected unit annual runoff from Type A/B soils and Type C/D soils using the curve number approach based on Technical Release Model 55 (TR-55) (USDA NRCS 2009), assuming turf grass is equivalent to open space in good condition (A/B CN = 59, C/D CN = 79) and adjusting for antecedent moisture condition. We then partitioned the CBWM unit runoff time series to Type A/B or Type C/D soil HRUs based on an adjustment ratio that considered both annual runoff (RO) and ratio of area of Type A/B (A_{ab}) to Type C/D (A_{cd}) soils in the Turfgrass class (mtg):

$$RO_{mtg} * A_{abcd} = (RO_{ab} * A_{ab}) + (RO_{cd} * A_{cd})$$

where RO = runoff per unit area (mgd/acre)

A = area (acres)

mtg = turfgrass HRU

abcd = all soil hydrologic groups

ab = soil hydrologic groups A + B

cd = soil hydrologic groups C + D

$$RO_{ab} * A_{ab} = (RO_{mtg} * A_{abcd}) - (RO_{cd} * A_{cd})$$

The ratio of unit runoff from soil hydrologic groups C + D to unit runoff from all soil groups was calculated based on estimated runoff from C + D using TR-55 and the unit runoff for turfgrass from the CBWM.

$$RO_{cd}/RO_{abcd} = RO_{cd,TR55}/RO_{mtg}$$

This was substituted into the previous equation to yield an estimate of remaining unit runoff attributable to soil hydrologic groups A + B:

$$RO_{ab} = (RO_{mtg} * A_{abcd}/A_{ab}) - (RO_{cd} * A_{cd}/A_{ab})$$

Adjusted unit recharge was calculated as $P - RO - PET$, where P = precipitation, RO = runoff, and PET = potential evapotranspiration.

Table 2. Translation of original Chesapeake Bay Watershed Model (CBWM) Hydrologic Response Units (HRUs) into WMOST HRUs based on aggregations within pervious and impervious classes, combination of pervious and impervious units, disaggregation by MS4 Permittee and hydrologic soil group. See Appendix C for CBWM class definitions.

Original CBWM land-use classes	Aggregated Land-use	Regulated Stormwater Permit	Soil Type	HRU	
				ID	Combined pervious/ impervious HRU
mir, mnr, mci	Impervious	Montgomery County	NA		
mch, mtg	Turfgrass	Montgomery County	A/B	1	MC Turfgrass with impervious A/B
mch, mtg	Turfgrass	Montgomery County	C/D	2	MC Turfgrass with impervious C/D
mir, mnr, mci	Impervious	City of Rockville	NA		
mch, mtg	Turfgrass	City of Rockville	A/B	3	RV Turfgrass with impervious A/B
mch, mtg	Turfgrass	City of Rockville	C/D	4	RV Turfgrass with impervious C/D
mir, mnr, mci	Impervious	MD State Highway Administration	NA		
mch, mtg	Turfgrass	MD State Highway Administration	A/B	5	MSHA Turfgrass with impervious A/B
mch, mtg	Turfgrass	MD State Highway Administration	C/D	6	MSHA Turfgrass with impervious C/D
mir, mnr, mci	Impervious	Other Regulated	NA		
mch, mtg	Turfgrass	Other Regulated	A/B	7	OReg Turfgrass with impervious A/B
mch, mtg	Turfgrass	Other Regulated	C/D	8	OReg Turfgrass with impervious C/D
for, hfr	Natural forested	NA	NA	9	Natural forested
osp, wto	Natural nonforested	NA	NA	10	Natural nonforested
wat	Water	NA	NA	11	Water

Calibration and Validation Approach

Although the CBWM is calibrated for the entire watershed, the HRU time series are generated at a relative coarse scale (CBWM land segment, approximately equivalent to Montgomery County). Thus, we cannot expect a perfect fit for the smaller CJC. For calibrations, we ran WMOST in simulation mode, with HRU areas fixed (min HRU area = max HRU area in Land Use Conservation table options; Appendix D. Screenshot D.1) and without any management options (stormwater BMPs, riparian restoration, alternative or structural BMPs) enabled. In practice, this is achieved by setting costs of all management actions to -9 within the model. This serves as a flag in the WMOST code to include or exclude various management practices from optimization equations. Although some stormwater control measures were in place in the CJC watershed in 2014, the CBWM outputs reflect those managed conditions, so this level of implementation was considered as zero added BMPs, rather than explicitly modelling these initial BMPs in WMOST. All drinking water demand in the CJC watershed is met through interbasin transfers so the number of water user classes was set to 1 and each element of the unaccounted water (UAW) and user demand and consumption time series was set to zero (Appendix D. Screenshot D.2). Virtually all CJC watershed is sewered so percentage public water users on septic was set to 0% (Appendix D. Screenshot D.3). Water demand could be set to zero since there is no public water utility in CJC, with drinking water supplied through interbasin transfers (Appendix D. Screenshot D.2). All lakes and wetlands were accounted for in the open water HRU with no downstream reservoir modelled. Private withdrawals and private discharge time series to surface water and groundwater were entered based on data obtained by MDE from withdrawal and discharge permits (Appendix D. Screenshot D.4). Infiltration and inflow to the sewer was set at 2.18 million gallons per day (MGD; City of Rockville Master Plan; <http://www.rockvillemd.gov/DocumentCenter/View/904> Appendix D. Screenshot D.5). Infrastructure modification options were set to -9 with associated costs set to zero (Appendix D. Screenshot D.6).

There are no USGS gaging stations in Cabin John Creek. Therefore, we calibrated WMOST for the pre-TMDL year of 2006 to observations from the nearby highly urbanized Anacostia Creek watershed at USGS gage 01650500 (Northwest Branch Anacostia River near Colesville, MD) and corrected for the ratio of Cabin John Creek: Anacostia watershed areas. The year 2006 had relatively average annual precipitation with the exception of two relatively large events (Figure 5). The initial “lumped” groundwater recession coefficient was calculated within WMOST based on HRU-specific groundwater recession coefficients from the CBWM, and then systematically altered to improve the fit between WMOST modelled and CJC-adjusted observations as determined by the Nash-Sutcliffe coefficient for streamflow (NSE, Moriasi et al. 2007).

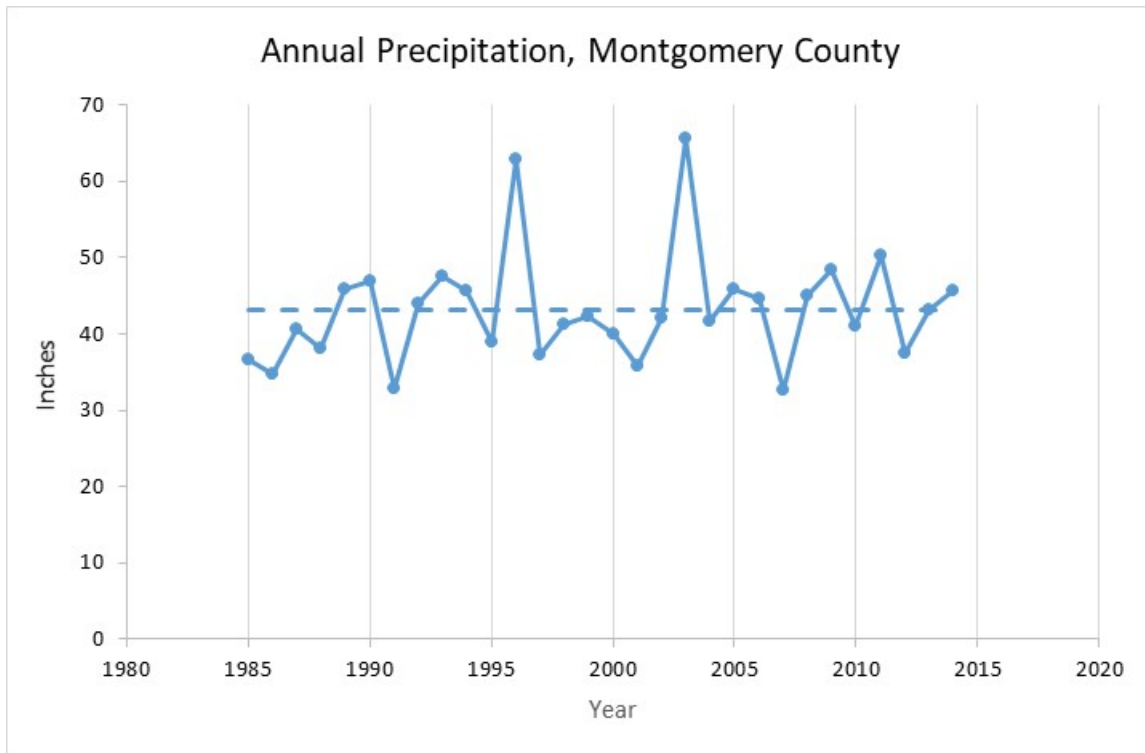


Figure 5. Annual (solid line) and average (dashed line) precipitation for Montgomery County. The year 2006 was an average precipitation year overall over the last 100 years but had the two highest 3-day precipitation totals in June.

Following calibration of the lumped groundwater recession coefficient, we calibrated the water quality loading time series based on observed versus modelled concentrations for total P and total N. Water quality observations were retrieved from the Maryland Ambient Water Quality Monitoring System (AWQMS; sample location in Figure 6). Suspended solids loading observations were very limited, so we also compared total annual average loads from WMOST (based on TR-55 adjusted CBWM outputs) with edge of stream loads generated by MDE for the sediment TMDL (MDE 2011) and used these ratios to correct load time series. Robustness of WMOST baseline results was checked using validation data for the year 2014 (starting point of optimization runs corresponding to end of Phase I WIPs).

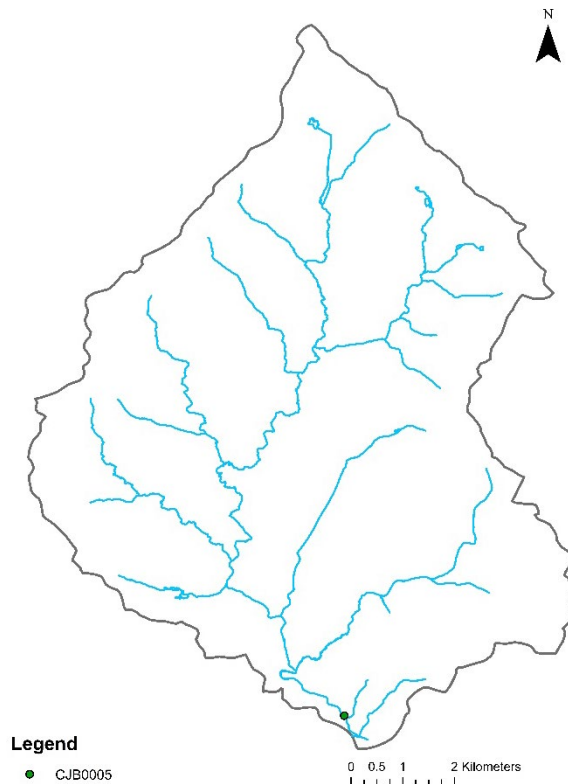


Figure 6. Location of water quality sampling station CJB0005 near mouth of Cabin John Creek.

Data Sources for Management Action Implementation Areas, Costs and Effectiveness

Riparian Buffer

We estimated three relative load groups for riparian buffer restoration measures using a modified version of the Chesapeake Bay Riparian Analysis Toolbox (<http://ches.communitymodeling.org/models.php>). The toolbox enables calculation of upland areas that drain to each riparian zone pixel (10 m segment). The toolbox was modified to allow calculation of flow pathways based on 10-meter digital elevation models (DEMs) from the beta high resolution NHD (1:24,000), using hydrologically-corrected DEMs (<https://www.usgs.gov/core-science-systems/ngp/national-hydrography/nhdplus-high-resolution>). Buffer zones were defined at 30-meter widths perpendicular to the stream channel (3 10-meter pixel depth). Upland HRU areas draining to each riparian pixel were converted to loads based on average annual unit HRU load. Riparian pixels in potential restoration areas were ranked by receiving load and then categorized into low, medium, and high relative load groups based on order-of-magnitude differences (Figure 7; Appendix D. Screenshot D.7). Riparian buffer percent load reductions were based on guidance from the Chesapeake Bay Program (Claggett and TetraTech 2014). Restoration costs were based on estimates from MD agencies compiled by MDE (Appendix E). Potential buffer restoration areas included turf grass and scrub

shrub, open mixed, and barren areas but not impervious area in developed HRUs, as removal of existing impervious areas is not necessarily practical and will significantly increase costs. Other potential factors influencing site selection for riparian restoration (e.g., those in the watershed resources registry; <https://watershedresourcesregistry.org/states/maryland.html>) were not considered but could be considered during implementation.

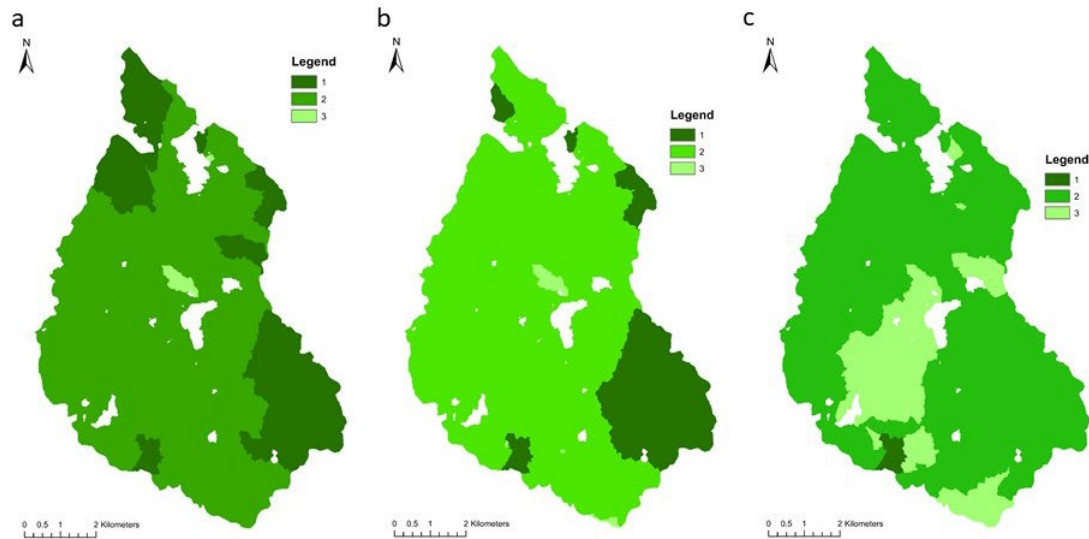


Figure 7. Relative load groups for contribution of upland HRUs to riparian buffer restoration areas for a) total N, b) total P, and c) total SS. Relative load group 1 has highest contributing load to riparian buffers. Most of the watershed falls in the intermediate relative load group. The highest relative load group overlaps for all three parameters in a small area in the SW portion of the watershed.

Stormwater Control Measures, Alternative, and Nonstructural BMPs

Level of implementation of existing BMPs by year was compiled by MDE [Appendix F]. We subtracted areas treated by existing BMPs in 2014 from total suitable HRU areas to obtain maximum HRU area available for treatment (Appendix D. Screenshot D.8). We calculated suitable areas for BMP placement based on siting criteria in MD stormwater manuals (CWP and MDE 2009) using the EPA Best Management Practices (BMPs) siting tool (<https://www.epa.gov/water-research/best-management-practices-bmps-siting-tool>; Table 3). MDE obtained estimates of installation costs of SCMs, alternative, and nonstructural BMPs from the City of Rockville, Montgomery County, and MD State Highway Administration (Appendix E). No operation and maintenance cost estimates were available from these agencies, so we used default WMOST estimates. Removal efficiencies for SCMs were calculated in WMOST via linkage to EPA's SUSTAIN tool; these removal efficiencies include effects of both runoff volume reductions and biogeochemical processes taking place within BMPs. In contrast, removal efficiencies currently used by the CBPO in assigning credits for SCMs are based on average reduction of runoff volume, and do not consider biogeochemical processes or interannual variability due to weather. Removal efficiencies for alternative and nonstructural BMPs were provided by MDE (MDE 2014; Appendix G).

Table 3. Siting criteria for stormwater control measures in MD and suitable acres for SCMs by HRU.

Stormwater Control Measures based on MDE naming criteria									
	Bioretention	Filtering Practices	Green Roof	Infiltration Practices	Permeable Pavement	Dry Detention Ponds & Hydrodynamic Structures, Dry Extended Detention Ponds, Dry Well, Stormwater to the Maximum Extent Practicable			
Stormwater Control Measures based on SUSTAIN categories									
Siting requirements based on MDE Stormwater Manual	Bioretention Basin (Acres)	Sand Filter (Acres)	Biofiltration w/UD (Acres)	Infiltration Basin (Acres)	Porous Pavement w/UD (Acres)	Extended Dry Detention Basin (Acres)	Dry Pond current implementation for Dry Pond to Wet Pond Conversion (Acres)	Wet Pond (Acres) Not modeled	
Soil Characteristics	Made Soil	OK	No criteria	f > 0.52 in/hr or A and B	A and B	A-D SUSTAIN			
Drainage Area (Acres)	< 5	< 10	No criteria	< 10	< 3 SUSTAIN	> 10 SUSTAIN			
Slope (%)	None	None	No criteria	< 15%	< 1%	< 15% SUSTAIN			
Ultra Urban > 80% imperviousness	OK	Depends	No criteria	Not Practical	OK	Not Practical			
HRU	Description	Suitable acres							
HRU1	Turfgrass A/B Montgomery County	674.96	1603.41	0	1440.79	767.42	22.99	787.22	9.20
HRU2	Turfgrass C/D Montgomery County	99.55	215.09	0	6.32	3.53	11.77	122.46	6.40
HRU3	Turfgrass A/B City of Rockville	121.02	387.99	0	369.13	229.87	3.35	63.78	0.80
HRU4	Turfgrass C/D City of Rockville	30.78	91.72	0	4.88	3.63	3.80	17.63	1.50
HRU5	Turfgrass A/B MD State Highway Administration	33.91	113.91	0	30.68	22.00	4.12	0	2.40
HRU6	Turfgrass C/D MD State Highway Administration	8.27	14.04	0	0.30	0.23	0.49	0	0.30
HRU7	Turfgrass A/B Other Regulated	14.61	74.91	0	70.08	47.99	0.78	0	0.14
HRU8	Turfgrass C/D Other Regulated	5.99	21.89	0	2.21	0.96	0.29	0	1.50

Sediment and Flow Loading Targets

The protocol for establishing annual suspended sediment loading targets for nontidal segments in the state of Maryland was developed in 2006 and amended in 2011 (MDE 2006, 2011). The methodology relies on calculation of the median long-term average load normalized to an all-forested condition associated with reference watersheds, originally calculated based on output from the Phase V CBWM (MDE 2006). Reference watersheds were defined based on the percentage of Maryland Biological Stream Survey monitoring stations, translated into watershed stream miles, that have an Index of Biotic Integrity Score (IBI) < 3 (MDE 2008). This methodology led to a sediment loading target defined as 3.3 times the sediment loading rate expected from a fully forested watershed.

Development of a total maximum daily load for Cabin John Creek required the translation of annual average loads into a maximum daily load. Following EPA methods (EPA 1991), the TMDL was estimated based on a 99th percentile probability of exceedance, the average annual sediment TMDL, and the coefficient of variation (CV) of the CJC reach simulation daily loads from the CBWM v5.2. MDE calculated a coefficient of variation of 9.4 from the modeled daily time series from 1985 – 2005. The maximum daily load for stormwater and nonpoint sources was estimated as the long-term average annual load multiplied by a factor that accounts for expected variability of daily loading values:

$$\text{MDL} = \text{LTA} * e^{(z\sigma - 0.5 \sigma^2)}$$

where

MDL = Maximum daily load

LTA = Long term average (average annual load)

Z = z-score associated with target probability level

$\sigma^2 = \ln(\text{CV}^2 + 1)$

CV = coefficient of variation based on arithmetic mean and standard deviation

The resulting dimensionless conversion factor from long term average annual loads is 14.7, with the daily load equivalent of 0.040 (14.7/365).

In evaluating different strategies for achieving the target annual and daily loads for CJC, we considered both a model-based approach (using our TR-55 adjusted Phase 6 CBWM inputs) and a flow-based empirical approach. The model-based approach assumes that the mechanistic representation of sediment sources, delivery, and transport in the CBWM is accurate. The flow-based empirical approach starts with the sediment loading target, then translates that into a flow target, using the empirical relationship between suspended sediment load and flow. Due to the lack of sufficient suspended sediment loading data for CJC, we determined the peak flows associated with sediment loading targets based on a sediment rating curve established for the Anacostia River sediment TMDL

(http://www.mde.maryland.gov/programs/Water/TMDL/ApprovedFinalTMDLs/Documents/www.mde.state.md.us/assets/document/AnacostiaSed_AppendixB_final.pdf). The

rationale for application of the flow-based empirical approach stems from Maryland's Biological Stressor Identification Analyses (BSIDs), which identify sediment/flow related stressors to in-stream habitat for aquatic life. These sediment related impacts to in-stream habitat are tied back to altered hydrologic regimes in Maryland's BSID reports. Therefore, it is assumed that by decreasing peak flow rates, sediment loads will also decrease, as will sediment impacts, to in-stream aquatic life.

Optimization Runs

Structural BMPs evaluated included Bioretention (SUSTAIN = Bioretention Basin), Filtering Practices (SUSTAIN = Sand Filter), Infiltration Practices (SUSTAIN = Infiltration Basin), Permeable Pavement (SUSTAIN = Porous Pavement w/UD), and Dry Detention Ponds (SUSTAIN = Extended Dry Detention Basin) (Appendix D. Screenshot D.9). We did not model wet pond creation because, given the siting criteria, there was virtually no suitable area left to implement additional wet ponds. Instead, we tested dry pond to wet pond conversions as a wet pond implementation using costs restricted to conversions only. Potential alternative BMPs tested included streambank stabilization/restoration (Appendix D. Screenshot D.10) and outfall enhancements (Appendix D. Screenshot. Nonstructural BMPs tested included street sweeping and tree canopy over turf (Appendix D. Screenshot D.11). The structural BMPs were tested with and without riparian buffer restoration (Table 4).

Table 4. Best management practices evaluated for Cabin John Creek watershed.

Structural BMPs	Alternative BMPs	Nonstructural BMPs
Bioretention	Streambank Stabilization/Restoration	Street Sweeping
Filtering Practices	Outfall Enhancement	Tree Canopy over Turf
Infiltration Practices	Riparian Buffer Restoration	
Permeable Pavement		
Dry Detention Ponds		
Dry Pond to Wet Pond Conversion		

We based initial optimization runs on 2014 weather conditions. The year 2014 had average precipitation levels overall with one large event. Estimated 2014 loads in the absence of post-2014 BMPs were used to establish target load reductions. We conducted an additional set of optimization runs using 2003 weather inputs. 2003 was a relatively wet year, although the 2003 maximum daily load actually was less than the 2014 maximum daily load due to the one large event in 2014. We included one set of water quality parameter constraints (TN or TP or TSS) per run, with both annual and maximum daily load targets specified. For TSS, we included the following constraints:

21% reduction of baseline annual load target plus 21% reduction of baseline maximum daily load (Method 1a) or

21% reduction of baseline annual load target plus maximum daily load target equal to $0.04 * \text{annual load target}$ (Method 1b)

The adjustment factor of 0.04 was derived by MDE as part of the CJC sediment TMDL and includes corrections for conversion between annual and daily time scales as well as adjustments for temporal variability (based on the coefficient of variation in a daily load time series) (MDE 2011).

We also ran optimizations using peak flow targets to control sediment loads. Method 2a involved finding representative daily discharge values corresponding to the 1-year 24-hour storm and establishing that as a peak flow target. The MDE stormwater manual establishes BMP designs to treat the 1-year 24-hour event. Method 2b involved estimating pre-development peak flow as a target by running WMOST with all HRUs converted to a forested state. Method 2c involved finding the peak flows associated with sediment loading targets based on a sediment rating curve established for the Anacostia River sediment TMDL

(http://www.mde.maryland.gov/programs/Water/TMDL/ApprovedFinalTMDLs/Documents/www.mde.state.md.us/assets/document/AnacostiaSed_AppendixB_final.pdf).

Results

Calibration

Flow

Results for the final 2006 WMOST hydrology calibration run are shown in Figure 8. The groundwater recession coefficient was adjusted to a value of 0.03. Calibration results were satisfactory, with a Nash Sutcliffe Efficiency (NSE) coefficient of 0.57 (satisfactory range $0.50 < \text{NSE} < 0.65$) (Moriassi et al. 2007), r^2 value of 0.59, and median bias of -1.9 (as compared to average measured flow of 34 cfs; Table 5). However, the adjusted CBWM results significantly underestimated runoff for the one very large event (189 mm precipitation) at the end of June.

Table 5. Fit statistics for WMOST calibration (2006) and validation (2014). Calibration was based on comparison to flow record from USGS gaging station 01650500 on the nearby highly urbanized Anacostia River, after correction for differences in watershed area.

Model run	Average Measured Flow	NSE Value	R ² Value	Relative Percent Error	Average Bias	Median Bias
Calibration	34.0	0.57	0.59	50.1	-0.6	-1.9
Validation	44.4	0.63	0.79	17.7	-5.4	-12.3

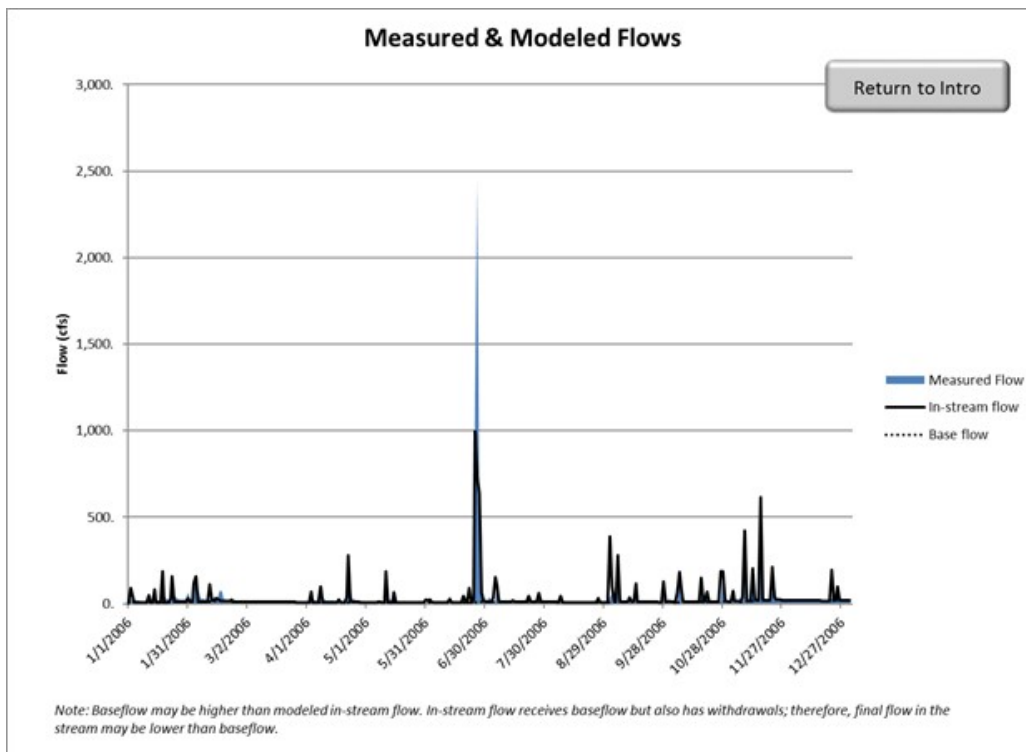


Figure 8. Results of WMOST 2006 model calibration for stream discharge (cfs).

Results of the model validation for baseline 2014 conditions are shown in Figure 9 and Table 5. The NSE is at the top end of the range of “satisfactory” NSE values based on Moriasi’s criteria (Moriasi et al. 2007), and near the lower end of the “good” range. Percent bias is also in the satisfactory range. As in 2006, modelled flows underestimated the largest peak flow in late April, with percent error approximately 3x the error associated with the peak June flow in 2006.

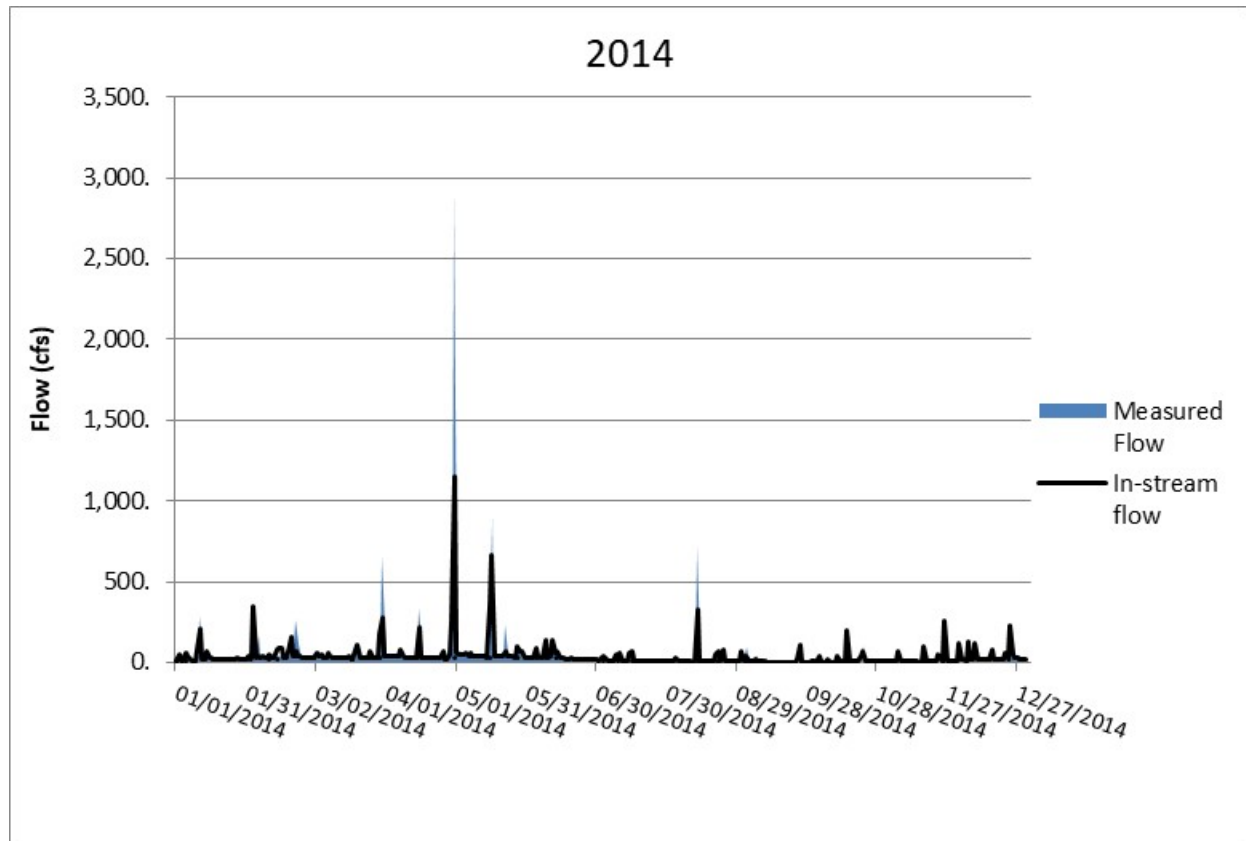


Figure 9. Results of WMOST validation procedure for modelled discharge for 2014 (pre-optimization).

TN

Results of the 2006 calibration run for TN are shown in Figures 10 and 11. Only twelve observations were available for comparison, and unfortunately, all were collected during baseflow conditions (Figure 10). The r^2 value was 0.61 with a percent bias of only -6.4% (considered very good), but the NSE was 0.25 (greater than zero so acceptable but less than satisfactory according to Moriasi et al. 2007). A comparison of modeled versus measured estimates of instantaneous TN load showed a negative bias for modeled loads at low values (intercept of -24 kg N/day) but a slope close to 1 (Figure 11). Observed loads were calculated as the product of instantaneous total nitrogen concentrations sampled near the outlet of Cabin John Creek (CBJ0005; Figure 11) and flow measurements from the USGS gaging station 01650500 in the nearby urbanized Anacostia watershed adjusted for differences in watershed area.

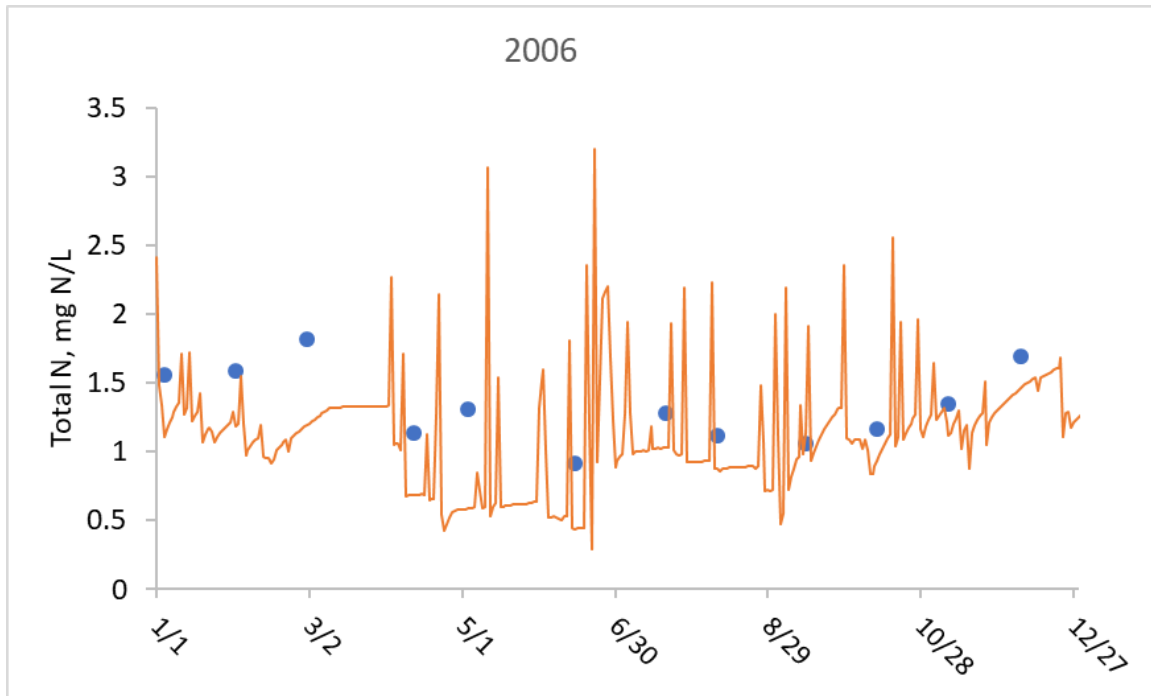


Figure 10. Results of 2006 calibration for total N, comparing WMOST measured (circles) and modeled values.

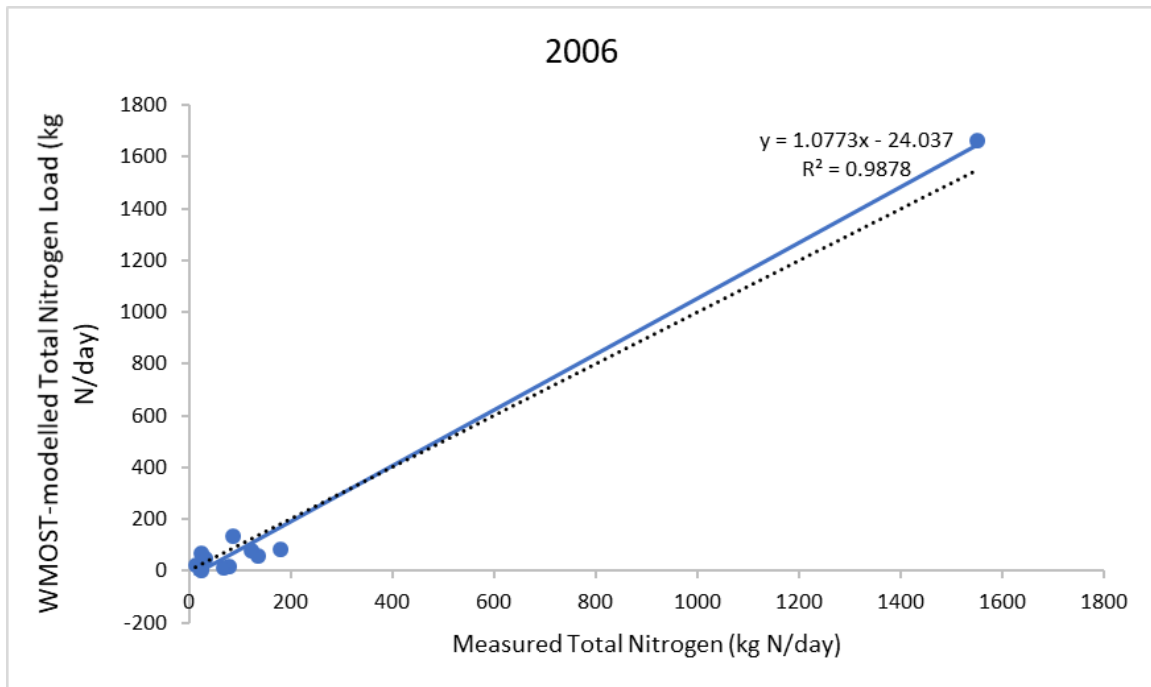


Figure 11. Comparison of measured vs modeled instantaneous total N load from WMOST for 2006 calibration period.

For the 2014 TN validation, however, the NSE was poor (-1.3) and the r^2 value was fair (0.62), with poor fits particularly for fall months (Figure 12).

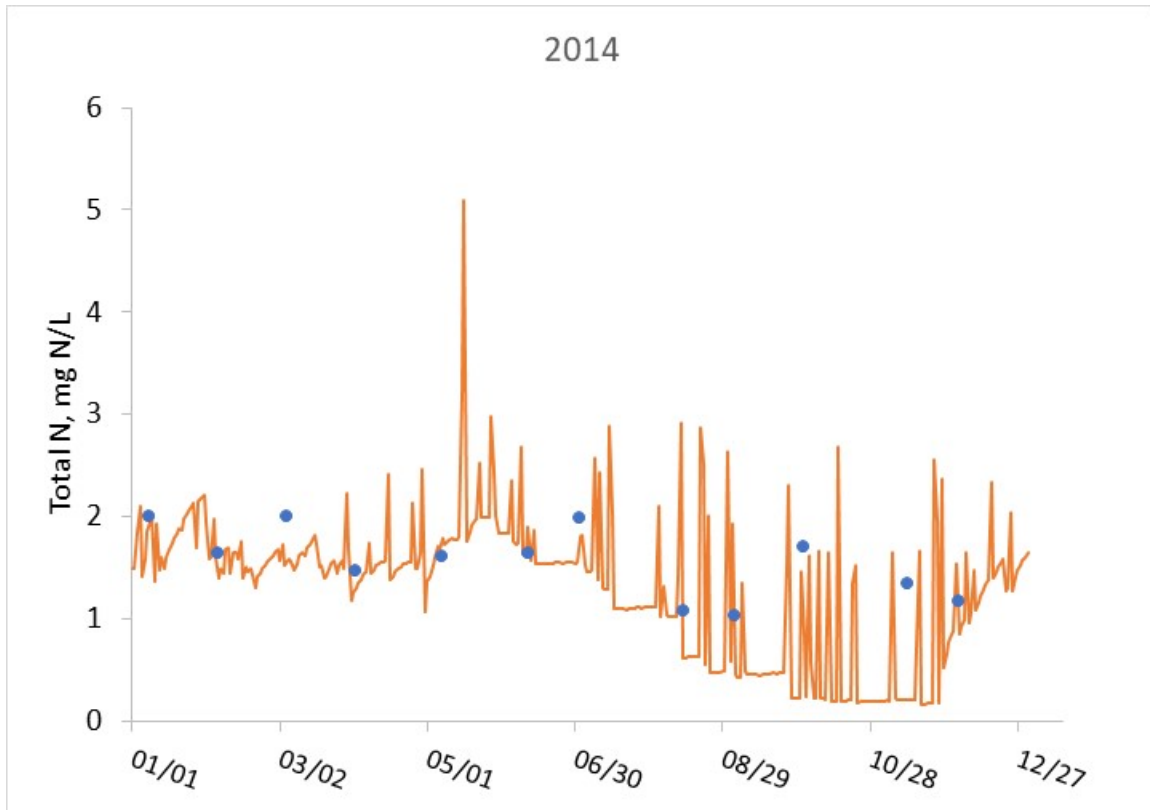


Figure 12. Comparison of WMOST validation: measured (circles) versus modeled concentration of TN in 2014.

TP

Results of the 2006 calibration run for TP are shown in Figure 13. Only twelve observations were available for comparison, and unfortunately, all were collected during baseflow conditions. Results of the model validation for baseline 2014 conditions are shown in Figure 14. Modeled values are generally greater than measured in spring through early summer but less than measured values in late summer through fall. Again, measured values were generally only available for baseflow conditions.

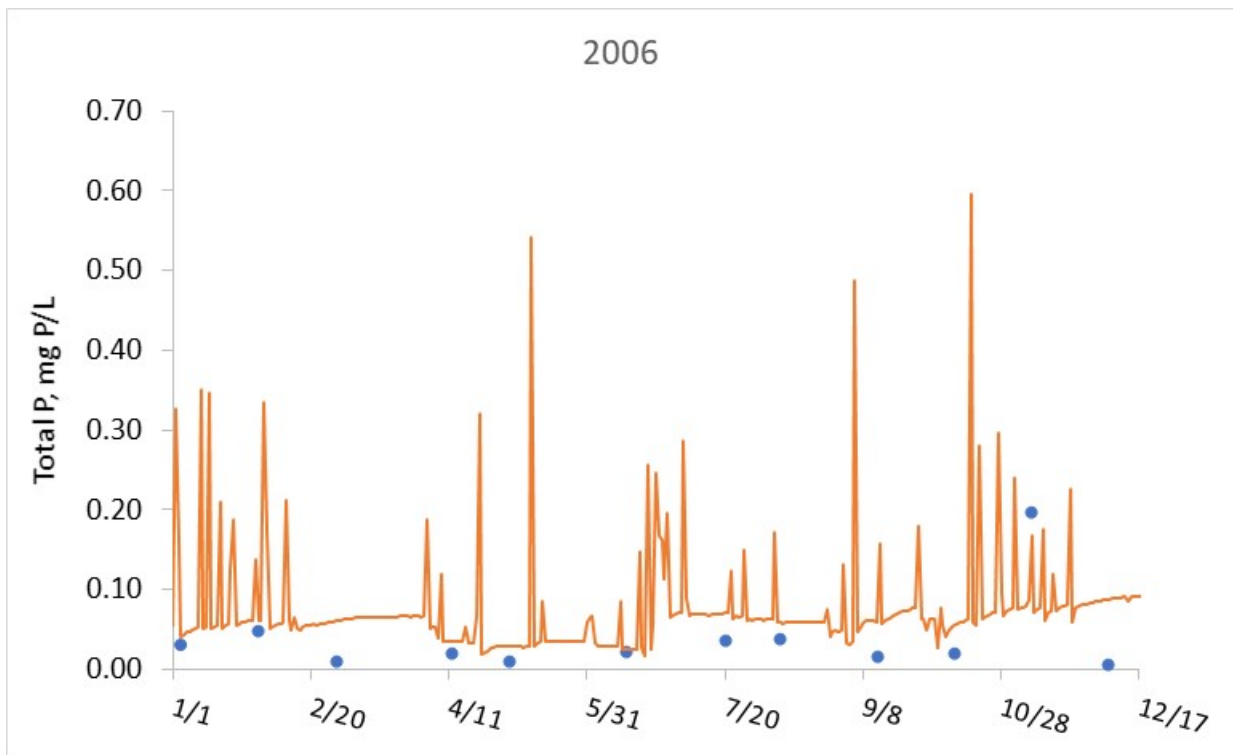


Figure 13. Calibration for TP in 2006, measured (circles) versus modeled TP.

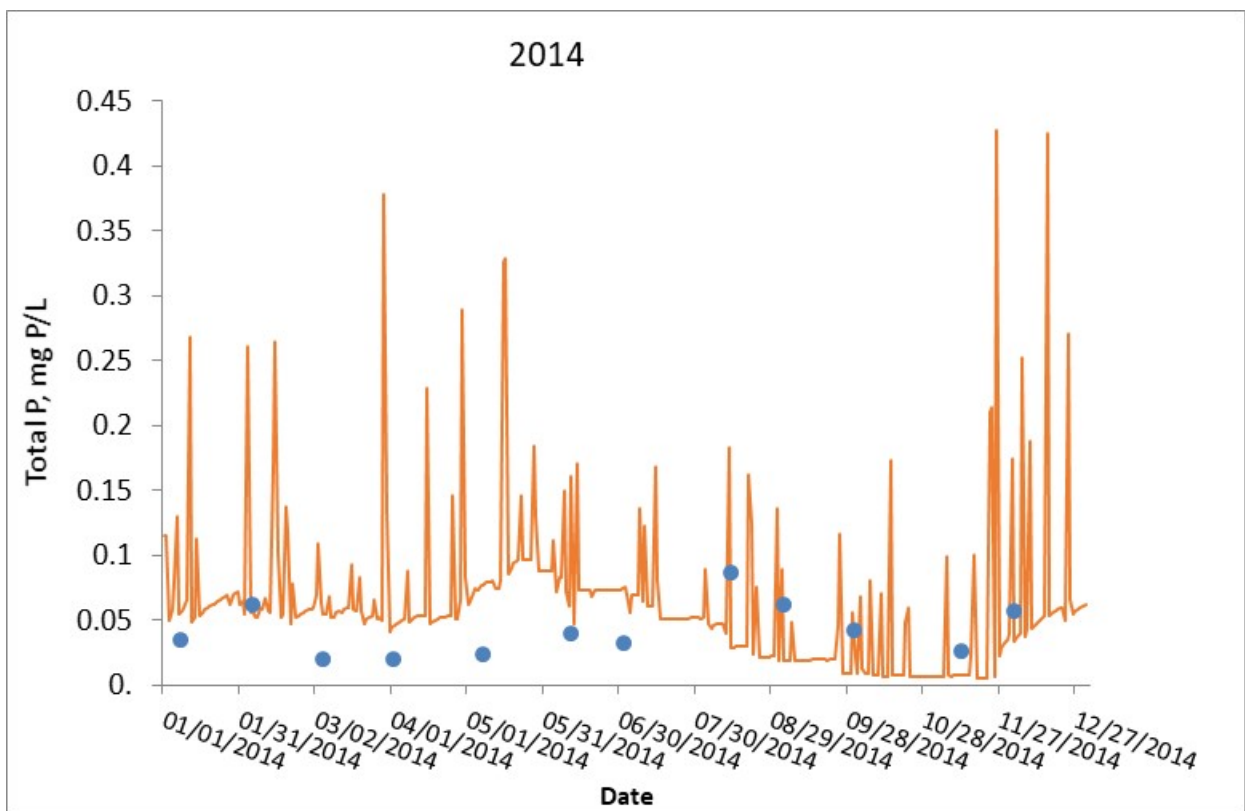


Figure 14. Validation of TP measured (circles) versus modeled concentrations in 2014.

TSS

Results of the 2006 calibration run for TSS are shown in Figure 15. Only six observations were available for comparison, and unfortunately, all were collected during baseflow conditions. Measured values were inadequate to calculate meaningful calibration statistics.

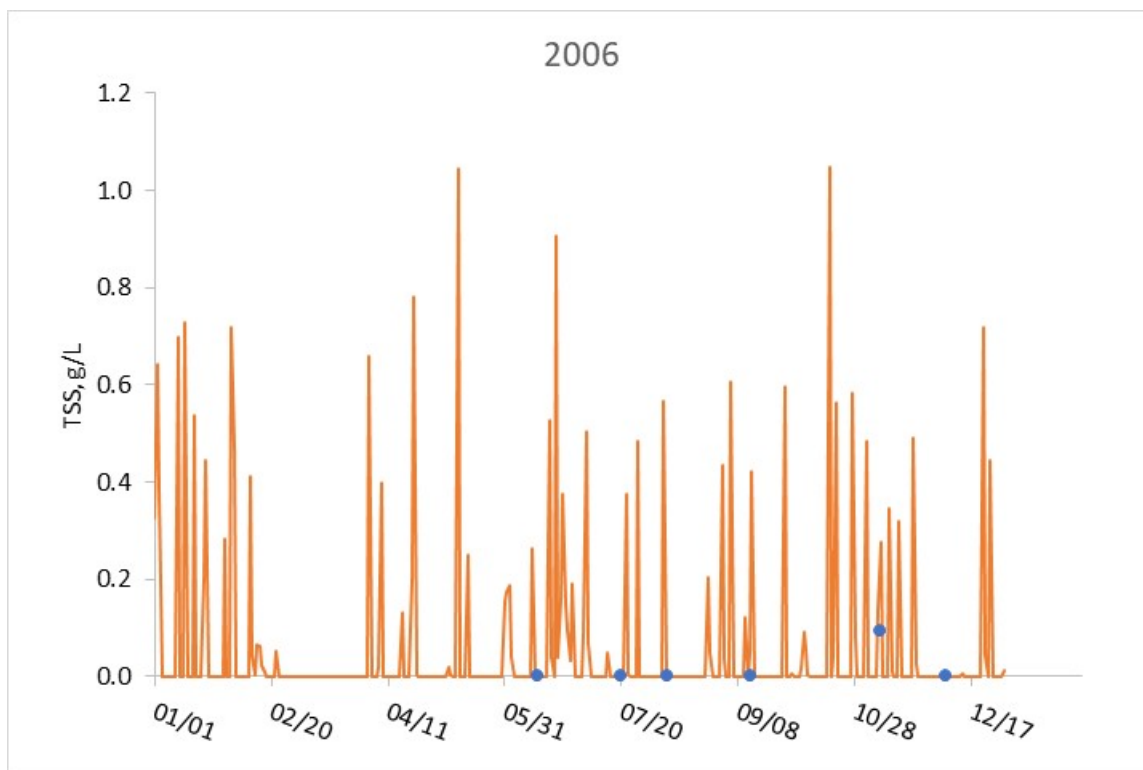


Figure 15. Time series of measured and modeled concentrations of total suspended sediment (g/L) for the calibration year, 2006.

Optimization Runs

Overall Rankings of Management Actions to Achieve Annual and Maximum Daily Load Targets

Table 6 provides the least-cost management options to achieve each of the parameter goals by year : 2014 baseline (average annual precipitation but one large event) and 2003 as a representative wet year (higher than average annual precipitation but with events more evenly distributed). For TN, the most cost-effective solution to achieve a 5% reduction in annual load and (2003 or 2014) maximum daily load included riparian buffer restoration for both 2014 (average/variable event) and 2003 (wet/even event) years. For TP, the most cost-effective solutions to achieve a 4% reduction in annual load and (2003 or 2014) maximum daily load also included riparian buffer. The least-cost solutions chosen to achieve a 21% reduction in annual and (2003 or 2014) maximum daily TSS loads included implementation of infiltration basins, dry pond to wet pond conversions and sand filters for both 2003 and 2014 weather regimes. When the MDE approach was used to derive a maximum daily load target that considers daily variation in

Table 6. Least-cost optimization selection of BMP management options for Cabin John Creek to meet maximum daily load and annual load targets for total N, total P, or total suspended solids under 2014 climate (average precipitation year with one large event) or 2003 (above average annual precipitation year). Riparian implementation area is area of riparian buffer restored. Implementation area for stormwater control measures (BMPs) represents area treated. First set of suspended sediment targets is based on 21% reduction of actual maximum daily load for baseline year, while second set of suspended sediment targets includes factor accounting for temporal variability of modelled maximum daily loads across years (MDE 2011). Colors correspond to BMP types.

All Management Options Considered	Max Daily Load Target (lb/day)	Max Annual Load Target (lb/year)	Target	Year	Selected Options	Max Daily Load (lb/day)	Annual Load (lb/year)	Daily Reduction (%)	Annual Reduction (%)	Implementation (Acres)	Sub-Cost (Millions \$)	Total Cost Sum (Millions \$)	Peak daily flow (cfs)	Peak daily flow (mgd)
None			N	2003	Baseline	11398	298694	--	--	--	--	--	1472	792
7 structural BMPs (Bioretention, sand filter, Infiltration Basin, Porous Pavement, Dry Detention Basin, Dry pond to Wet pond conversion) + riparian buffers	10830	277471	N	2003	riparian buffers	10798	281668	5.26%	5.70%	74		0.514	940	506
None			N	2014	Baseline	18260	146456	--	--	--	--	--	1386	746
riparian buffers	17347	139133	N	2014	riparian buffers	16936	138869	7.25%	5.18%	114		0.914	1309	705
None			P	2003	Baseline	1012	17198	--	--	--	--	--	969	522
riparian buffers	972	16510	P	2003	riparian buffers	958	16326	5.30%	5.07%	24		0.166	948	510
None			P	2014	Baseline	1178	10333	--	--	--	--	--	1387	746
riparian buffers	1130	9920	P	2014	riparian buffers	1121	9890	4.84%	4.29%	24		0.166	1327	714
None			TSS	2003	Baseline	779657	12886780	--	--	--	--	--	1472	792
7 structural BMPs (Bioretention, sand filter, Infiltration Basin, Porous Pavement, Dry Detention Basin, Dry pond to Wet pond conversion)	615929	10180556	TSS	2003	2.75 in Infiltration Basin + Dry Pond to Wet Pond Conversion + 2.75" Sand Filter	615928	10173479	21.00%	21.05%	1893 + 982 + 92	18.42 + 3.76 + 1.14	13.32	1271	684

All Management Options Considered	Max Daily Load Target (lb/day)	Max Annual Load Target (lb/year)	Target	Year	Selected Options	Max Daily Load (lb/day)	Annual Load (lb/year)	Daily Reduction (%)	Annual Reduction (%)	Implementation (Acres)	Sub-Cost (Millions \$)	Total Cost Sum (Millions \$)	Peak daily flow (cfs)	Peak daily flow (mgd)
None			TSS	2014	Baseline	662371	9305934	--	--	--	--		2059	1108
7 structural BMPs (Bioretention, sand filter, Infiltration Basin, Porous Pavement, Dry Detention Basin, Dry pond to Wet pond conversion)	523273	7351688	TSS	2014	2.75 in Infiltration Basin + Dry Pond to Wet Pond Conversion + 2.75" Sand Filter	523272	7318198	21.00%	21.36%	1924 + 991 + 81	18.79 + 3.76 + 1.00	23.55	1789	963
None			TSS	2003	Baseline	779657	12886780	--	--	--	--		1472	792
7 structural BMPs (Bioretention, sand filter, Infiltration Basin, Porous Pavement, Dry Detention Basin, Dry pond to Wet pond conversion)	407,222.24	10,180,555.97	TSS	2003	2.75 in Infiltration Basin + Dry Pond to Wet Pond Conversion + 2.75" Sand Filter + 2.75" Bioretention Basin + Riparian Buffer Restoration	407222	6799471	47.77%	47.24%	1924 + 991 + 1893 + 65 + 393	18.79 + 3.76 + 24.65 + 1.74 + 1.37	50.31	1076	579
None			TSS	2014	Baseline	662371	9305934	--	--	--	--		2059	1108
7 structural BMPs (Bioretention, sand filter, Infiltration Basin, Porous Pavement, Dry Detention Basin, Dry pond to Wet pond conversion)	294,067.51	7,351,687.80	TSS	2014	2.75 in Infiltration Basin + Dry Pond to Wet Pond Conversion + 2.75" Sand Filter + 2.75" Bioretention Basin + Riparian Buffer Restoration	294067	4095558	55.60%	55.99%	1924 + 991 + 2463 + 817 + 393	18.79 + 3.76 + 44.64 + 22.22 + 1.37	90.78	1410	759

TSS load, additional BMPs and level of implementation were required – a combination of infiltration basins, dry pond to wet pond conversions, sand filters, bioretention basins, and riparian buffer restoration.

Management Actions Required to Achieve Flow Targets Consistent with 21% Sediment TMDL Target

Achieving the flow target associated with the 21% reduction in MDE-estimated maximum daily loads requires the greatest level of BMP implementation. A flow target based on estimates of pre-development 1 yr 24-hour events (11 - 13 cfs) was not achievable with any level of BMP implementation. Peak flow targets based on application of sediment rating curves (499 cfs in 2014, 535 cfs in 2003) were also not achievable. The minimum achievable targets based on incremental scenarios of steadily decreasing peak flow targets were 700 cfs for the 2014 weather regime and 525 cfs for the 2003 weather regime. These flow targets required implementation of a combination of sand filters, infiltration basins, and porous pavement for both 2003 and 2014 weather regimes (Figure 16). However, peak flows for least-cost solutions for 21% reduction of TSS annual loads described above were similar to the minimum peak flows possible (Table 6).

The CBWM is designed to model edge of stream loads but may not fully account for effects of peak flows on suspended sediment and bedload derived from stream bank erosion. The latter are captured by sediment rating curves, i.e., the relationship between discharge and suspended sediment load. Analyses performed by MDE for the Anacostia River sediment TMDLs using sediment rating curves suggested that 75% of increased sediment loads over the last 65 years were due to increases in peak flow events. Although WMOST identified riparian buffer restoration as least-cost solutions to achieve load reductions for TN and TP, these solutions only achieved 2-6% reductions in maximum daily flows.

TN and TP Scenarios with TSS Optimal Management

The effects of optimal management solutions to meet 21% reductions in annual and maximum daily TSS loads on TN and TP loads were evaluated by fixing these management solutions as scenarios and running WMOST for TN and TP but without targets. Optimal management for sediment (including infiltration basins, dry to wet pond conversions, and sand filters) would achieve annual load reductions for TN of 12.5 – 12.6% (Figure 17) and annual load reductions for TP of 15.5 – 15.6% (Figure 18). Even greater reductions are predicted for maximum daily loads, of 17.4 – 17.9% for TN and 16.6 – 19.9% for TP (Table 7). Peak daily flows would be reduced by 13 – 13.6% (Figure 16).

Table 7. Effect of applying optimal strategies for reducing TSS loads by 21% on achievement of daily maximum and annual loading goals for total N and total P.

All Management Options Considered	Max Daily Load Target (lb/day)	Max Annual Load Target (lb/year)	Target Year	Model Run	Max Daily Load (lbs)	Annual Load (lbs)	Daily Reduction (%)	Annual Reduction (%)	Implementation (Acres)	Total Cost (Millions \$)	Total Cost Sum (Millions \$)	Peak daily flow (cfs)	Peak daily flow (mgd)
Simulate optimal solution for TSS control	10830	277471	N	2003	2.75 in Infiltration Basin + Dry Pond to Wet Pond Conversion + 2.75" Sand Filter	9417	261200	17.38%	12.55%	2,976.57	23.64		452.00
Simulate optimal solution for TSS control			P	2003	2.75 in Infiltration Basin + Dry Pond to Wet Pond Conversion + 2.75" Sand Filter	811	14539	19.88%	15.46%	2,976.57	23.64		452.00
Simulate optimal solution for TSS control	17346.62	139133.5	N	2014		14994.14	128177.5	17.88%	12.48%	2996.847	23.87769		652.4349
Simulate optimal solution for TSS control			P	2014		982.4265	8720.247	16.57%	15.61%	2996.847	23.87769		652.4349

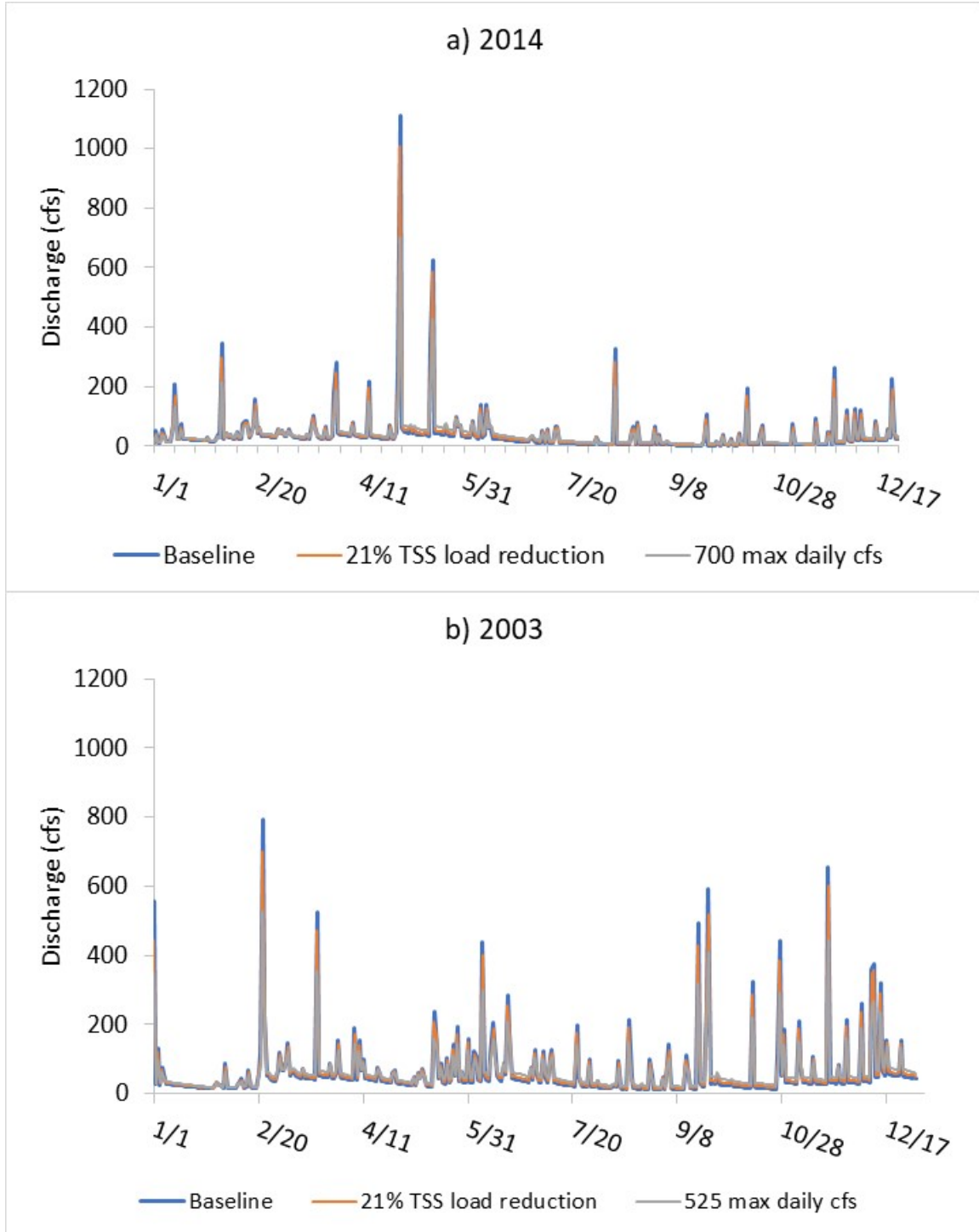


Figure 16. Flow time series associated with baseline discharge, optimization for 21% TSS load reduction, and minimum peak flow solutions for a) 2014 and b) 2003 conditions.

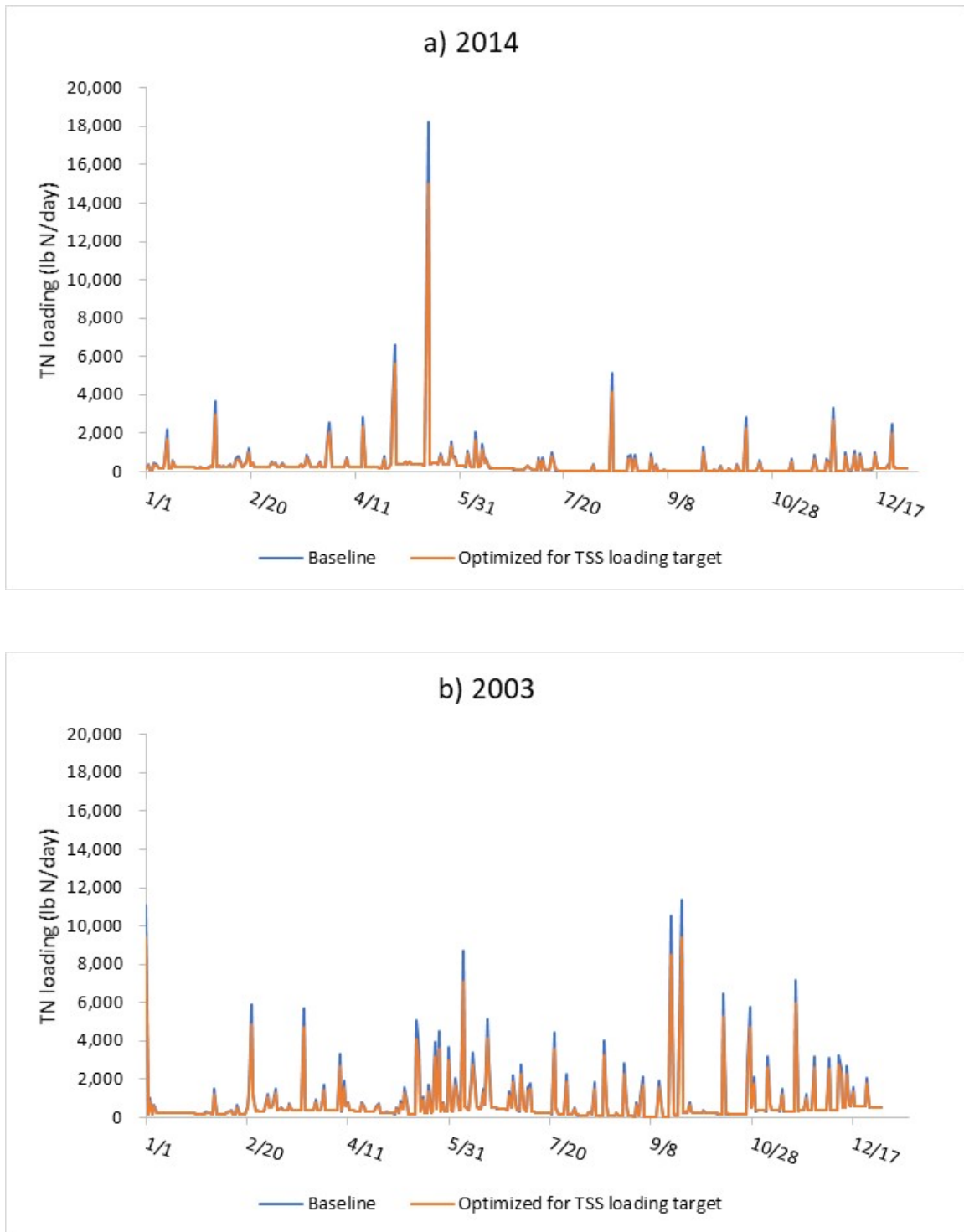


Figure 17. Daily loads for total N (TN) under optimal management regime for meeting 21% sediment load reduction under a) 2014 weather conditions and b) 2003 conditions.

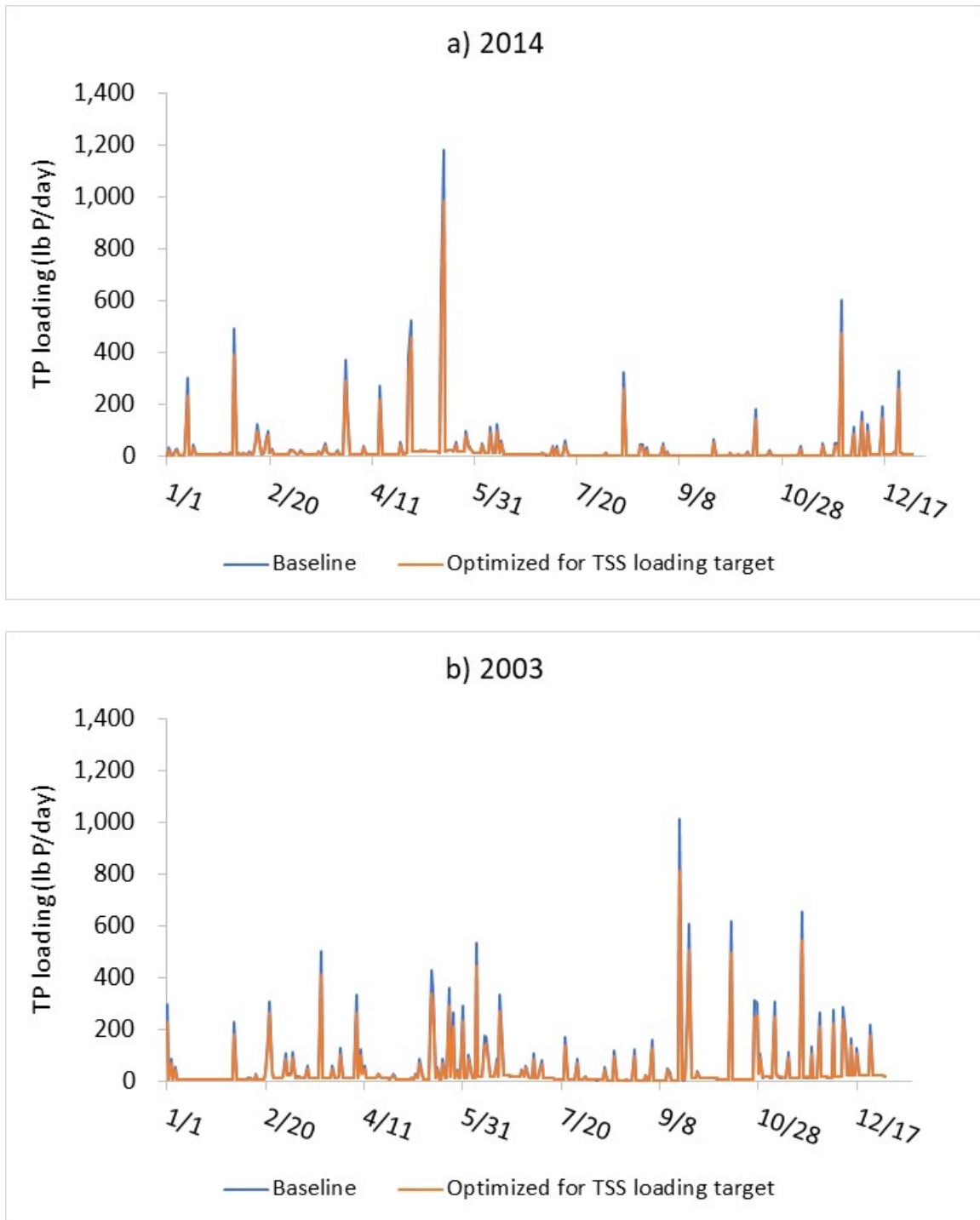


Figure 18. Daily loads for total phosphorus (TP) under optimal management regime for meeting 21% sediment load reduction under a) 2014 weather conditions and b) 2003 conditions.

Comparison of Stormwater BMP and Riparian Implementation by HRU

Optimal solutions for 21% reductions in annual and maximum daily loads of TSS required full implementation of infiltration basins on all suitable land and full conversion of existing dry ponds to wet ponds across all permittee types, supplemented by an implementation of sand filters to treat 81 acres, a small fraction of the total area that could be treated by sand filters. Site constraints limited implementation based on soil type (infiltration basins) and drainage area (infiltration basins and dry detention pond to wet pond conversions), but not for sand filters (Table 3). Sand filters were preferentially allocated to Montgomery County jurisdictions with Soil Hydrologic Groups Type A or B (Table 8).

Table 8. Optimal treatment of HRU acres by stormwater control measures for 2014 vs 2003 input conditions to meet 21% TSS load reduction.

HRU ¹	Base-line Area (acres)	EIA ²	INFIL ³	New Implementation		New Implementation + Conversions		
				2.75 in Infiltration Basin	2.75 in Sand Filter	2.75 in Infiltration Basin	Dry Pond to Wet Pond Conversion	2.75 in Sand Filter
2014 inputs								
Tg A/B MC	9,073	0.31	0.4	1441	937	1441	787	81
Tg C/D MC	1,465	0.31	0.05	6	0	6	122	0
Tg A/B Rv	1,544	0.37	0.4	369	0	369	64	0
Tg C/D Rv	372	0.37	0.05	5	0	5	18	0
Tg A/B SHA	445	0.68	0.4	31	0	31	0	0
Tg C/D SHA	130	0.68	0.05	0	0	0	0	0
Tg A/B Oth	333	0.23	0.4	70	0	70	0	0
Tg C/D Oth	122	0.23	0.05	2	0	2	0	0
2003 inputs								
Tg A/B MC	9,073	0.31	0.4	1441	919	1441	787	92
Tg C/D MC	1,465	0.31	0.05	6	0	6	122	0
Tg A/B Rv	1,544	0.37	0.4	369	0	369	64	0
Tg C/D Rv	372	0.37	0.05	5	0	5	18	0
Tg A/B SHA	445	0.68	0.4	31	0	0	0	0
Tg C/D SHA	130	0.68	0.05	0	0	0	0	0
Tg A/B Oth	333	0.23	0.4	70	0	70	0	0
Tg C/D Oth	122	0.23	0.05	2	0	2	0	0 ⁴

¹ HRU = Hydrologic Response Unit, Tg = turfgrass + impervious, A+B = soil hydrologic groups A + B, C + D = soil hydrologic groups C + D, MC = Montgomery County, Rv = Rockville, SHA = State Highway Administration, Oth = Other regulated entity

² EIA = effective impervious area

³ INF = infiltration capacity

Under the more stringent requirements imposed by the MDE sediment TMDL targeting approach (Method 1b), greater implementation of sand filters was required for 2014 conditions with preference for jurisdictional implementation by Montgomery County > City of Rockville > State Highway Administration > Other Regulated entities. Within each jurisdiction, preference was given for implementation on Type A or B soils as compared to Type C or D soils. Riparian buffer restoration recommendations were more evenly split between Soil Hydrologic Groups A + B as compared to groups C + D (Table 9). Under the even more stringent requirements based on peak flow targets only, preference was given to porous pavement implementation as a third option as compared to conversion of dry ponds to wet ponds and addition of bioretention basins or riparian restoration.

Table 9. Optimal treatment of HRU acres to meet 21% annual and maximum daily load reductions for TSS, including MDE factor for temporal variability.

HRU ¹	2.75 in Infiltration Basin	Dry Pond to Wet Pond Conversion	2.75 in Sand Filter	2.75 in Bioretention Basin	Riparian Buffer Restoration
2014 conditions					
Tg A/B MC	1441	787	1603	675	133
Tg C/D MC	6	122	215	100	135
Tg A/B Rv	369	64	388	0	33
Tg C/D Rv	5	18	92	0	23
Tg A/B SHA	31	0	114	34	5
Tg C/D SHA	0	0	14	8	5
Tg A/B Oth	70	0	36	0	14
Tg C/D Oth	2	0	0	0	23
Natural-nonf	0	0	0	0	20
2003 conditions					
Tg A/B MC	1441	787	1603	0	133
Tg C/D MC	6	122	215	65	135
Tg A/B Rv	369	64	0	0	33
Tg C/D Rv	5	18	0	0	23
Tg A/B SHA	31	0	0	0	5
Tg C/D SHA	0	0	0	0	5
Tg A/B Oth	70	0	75	0	14
Tg C/D Oth	2	0	0	0	23
Natural-nonf	0	0	0	0	20

¹ HRU = Hydrologic Response Unit, Tg = turfgrass + impervious, A+B = soil hydrologic groups A + B, C + D = soil hydrologic groups C + D, MC = Montgomery County, Rv = Rockville, SHA = State Highway Administration, Oth = Other regulated entity

Optimal solutions for TN or TP reductions alone differed from optimal solutions for TSS reductions. If only new stormwater control measures were considered with no riparian buffer restoration, infiltration basins alone would be the most cost-effective option to meet TN or TP load reductions required for the Chesapeake Bay TMDL, implemented on Montgomery County, Rockville, or Other Regulated jurisdictions for TN and on Montgomery County and Other Regulated for TP. If retrofits are considered, then conversion of existing extended detention dry ponds to wet ponds would be the least-cost solution, with implementation in Montgomery County and Other Regulated jurisdictions (Tables 10, 11). In general, placement of infiltration basins on Type A/B soils was favored while retrofits to wet ponds were favored on Type C/D soils.

Table 10. Optimal treated acres of HRUs by BMPs for TN load reductions considering 1) all stormwater control measures (SCM), 2) all SCMs plus retrofits, or 3) riparian buffer restoration.

HRU ¹	Baseline Area (acres)	EIA ²	INFIL ³	New	Retrofit	Riparian buffer
				2.75 in Infiltration Basin	Dry Pond to Wet Pond Conversion	
2014 inputs						
Tg A/B MC	9,073	0.31	0.4	33	0	10
Tg C/D MC	1,465	0.31	0.05	0	478	55
Tg A/B Rv	1,544	0.37	0.4	357	0	14
Tg C/D Rv	372	0.37	0.05	0	0	9
Tg A/B SHA	445	0.68	0.4	0	0	4
Tg C/D SHA	130	0.68	0.05	0	0	1
Tg A/B Oth	333	0.23	0.4	333	333	6
Tg C/D Oth	122	0.23	0.05	122	74	9
Natural-nonf	0	0	0	0	0	8
2003 inputs						
Tg A/B MC	9,073	0.31	0.4	366	0	10
Tg C/D MC	1,465	0.31	0.05	0	294	55
Tg A/B Rv	1,544	0.37	0.4	0	0	6
Tg C/D Rv	372	0.37	0.05	0	0	3
Tg A/B SHA	445	0.68	0.4	0	0	0
Tg C/D SHA	130	0.68	0.05	0	0	0
Tg A/B Oth	333	0.23	0.4	333	333	0
Tg C/D Oth	122	0.23	0.05	0	122	0
Natural-nonf	0	0	0	0	0	0

¹ HRU = Hydrologic Response Unit, Tg = turfgrass + impervious, A+B = soil hydrologic groups A + B, C + D = soil hydrologic groups C + D, MC = Montgomery County, Rv = Rockville, SHA = State Highway Administration, Oth = Other regulated entity

² EIA = effective impervious area

³ INF = infiltration capacity

Table 11. Optimal treated acres of HRUs by BMPs for TP load reductions considering 1) all stormwater control measures (SCM), 2) all SCMs plus retrofits, or 3) riparian buffer restoration.

HRU ¹	Baseline Area (acres)	EIA ²	INFIL ³	New	Retrofit	Riparian buffer
				2.75 in Infiltration Basin	Dry Pond to Wet Pond Conversion	
2014 inputs						
Tg A/B MC	9,073	0.31	0.4	377	0	10
Tg C/D MC	1,465	0.31	0.05	0	403	14
Tg A/B Rv	1,544	0.37	0.4	0	0	0
Tg C/D Rv	372	0.37	0.05	0	0	0
Tg A/B SHA	445	0.68	0.4	0	0	0
Tg C/D SHA	130	0.68	0.05	0	0	0
Tg A/B Oth	333	0.23	0.4	333	333	0
Tg C/D Oth	122	0.23	0.05	0	0	0
Natural-nonf	0	0	0	0	0	0
2003 inputs						
Tg A/B MC	9,073	0.31	0.4	384	0	10
Tg C/D MC	1,465	0.31	0.05	0	298	14
Tg A/B Rv	1,544	0.37	0.4	0	0	0
Tg C/D Rv	372	0.37	0.05	0	0	0
Tg A/B SHA	445	0.68	0.4	0	0	0
Tg C/D SHA	130	0.68	0.05	0	0	0
Tg A/B Oth	333	0.23	0.4	333	333	0
Tg C/D Oth	122	0.23	0.05	0	122	0
Natural-nonf	0	0	0	0	0	0

¹ HRU = Hydrologic Response Unit, Tg = turfgrass + impervious, A+B = soil hydrologic groups A + B, C + D = soil hydrologic groups C + D, MC = Montgomery County, Rv = Rockville, SHA = State Highway Administration, Oth = Other regulated entity

² EIA = effective impervious area

³ INFIL = infiltration capacity

Alternative and Nonstructural BMPs

Many of the alternative and nonstructural BMPs proved to be infeasible to meet maximum daily and/or annual loads. Neither street sweeping, nor streambank stabilization yielded feasible solutions for TN or TP in 2003 or 2014. Tree canopy implementation could meet goals but at a much higher cost as compared to either stormwater control measures or riparian restoration. Neither tree canopy, street sweeping, nor streambank stabilization could meet TSS load reduction targets for 2003 or 2014 conditions (Table 12).

Distribution of Proposed Riparian Zone Restoration Among Relative Load Groups

Required riparian buffer restoration to meet TN or TP load reductions did not differ between 2014 and 2003 conditions. Greater restoration would be required to meet TN load reductions, split mainly between relative load groups 1 and 2, as compared to restoration to meet TP load reductions, from relative load group 1 on Montgomery County jurisdiction only (Table 13). Riparian buffer removal efficiencies are twice as great for TP as for TN (Appendix G).

Table 12. Least-cost optimization selection of alternative and nonstructural BMP management options for Cabin John Creek to meet maximum daily load and annual load targets for total N, total P, or total suspended solids under 2014 climate (average precipitation year with one large event) or 2003 (above average annual precipitation year). STSW = street sweeping (100x/yr), TC = tree canopy, STST = streambank stabilization. Inf = infeasible.

Parameter	Options	Max Daily Load Target (lb/day)	Max Annual Load Target (lb/year)	Target	Year	Max Daily Load (lb/day)	Annual Load (lb/year)	Daily Reduction (%)	Annual Reduction (%)	Implementation (Acres)	Total Cost (Millions \$)	Peak Daily flow (cfs)
TN	None			N	2003	11398	298694	--	--	--	--	807
TN	STSW	10830	277471	N	2003						Inf	
TN	TC	10830	277471	N	2003	8939	249751	21.58%	16.39%	7180	175	807
TN	STST	10830	277471	N	2003						Inf	
TN	None			N	2014	18260	146456	--	--	--	--	1154
TN	STSW	17347	139133	N	2014	--	--	--	--	--	Inf	
TN	TC	17347	139133	N	2014	14299	125385	21.69%	14.39%	7180	175	1154
TN	STST	17347	139133	N	2014						Inf	
TP	None			P	2003	1012	17198	--	--	--	--	807
TP	STSW	972	16510	P	2003						Inf	
TP	TC	972	16510	P	2003	784	14284	22.50%	16.95%	7180	175	807
TP	STST	972	16510	P	2003						Inf	
TP	None			N	2014	1178	10333					1154
TP	STSW	1130	9920	N	2014						Inf	
TP	TC	1130	9920	N	2014	933	8526	20.74%	17.49%	7180	175	1155
TP	STST	1130	9920	N	2014						Inf	
TSS	None			TSS	2003	779657	12886780					807
TSS	STSW	615,929	10,180,556	TSS	2003						Inf	
TSS	TC	615,929	10,180,556	TSS	2003						Inf	
TSS	STST	615,929	10,180,556	TSS	2003						Inf	
TSS	None			TSS	2014	662371	9305934					1154
TSS	STSW	523,273	7,351,688	TSS	2014						Inf	
TSS	TC	523,273	7,351,688	TSS	2014						Inf	
TSS	STST	523,273	7,351,688	TSS	2014						Inf	

Table 13. Required conversion of HRUs to forested riparian buffer by relative load group for TN or TP load reductions to meet Chesapeake Bay TMDL targets at least-cost.

HRU conversions	Relative load group					
	1		2		3	
	TN	TP	TN	TP	TN	TP
Tg A/B MC	10.41	9.55	0	0	0.02	0
Tg C/D MC	13.07	14.39	41.54	0	0.05	0
Tg A/B Rv	6.26	0	7.10	0	0.20	0
Tg C/D Rv	2.75	0	6.32	0	0	0
Tg A/B SHA	0.96	0	1.02	0	0	0
Tg C/D SHA	0.76	0	0.00	0	0	0
Tg A/B Oth	0.58	0	5.15	0	0.02	0
Tg C/D Oth	0.72	0	8.00	0	0.47	0
Natural-nonf	0.75	0	6.88	0	0.61	0

Discussion

Model Fits

In this application of WMOST, we used HRU-specific unit hydrology (runoff, recharge) and pollutant loads from the Phase 6 CBWM. Time series were subsequently modified using TR-55 to partition hydrology and loads between HRUs on soil hydrologic groups A+B and groups C+D. Specifically, contributions from soil groups C+D were estimated using TR-55 and subtracted from contributions from all soil groups to estimate contributions from soil groups A+B to ensure that total loads per original HRU remained the same. The CBWM was calibrated to USGS Weighted Regressions on Time, Discharge and Season (WRTDS) loads by the Chesapeake Bay Program Office using observations from watersheds ranging in area from 7.4 mi² to 27,086 mi² (ftp://ftp.chesapeakebay.net/Modeling/Phase6/Phase_6_201710/Watershed_Model/WSM_Outputs/WRTDS_Comparison/). Model fits for the CBWM overall were excellent (TN NSE = 0.998, TP NSE = 0.997, TSS NSE = 0.987). Estimated loads per unit area were also very good (TN NSE = 0.955, TSS NSE = 0.965, TP NSE = 0.966). A comparison of modelled vs estimated WRTDS loads for the NW Branch of the Anacostia River showed modeled values that were 98.6 – 98.9% of estimated loads. However, model fits for concentration were highly variable and worse than load comparisons for the NW Branch of the Anacostia ($r^2 = 0.14$ for TN concentrations; $r^2 = 0.72$ for TN loads¹; $r^2 = 0.55$ for TP concentrations², $r^2 = 0.72$ for TP loads; $r^2 = 0.36$ for TSS concentrations, $r^2 = 0.55$ for TSS loads³). Given the goal of optimizing the CBWM for the entire basin, with a greater emphasis on load estimation, it is not surprising that modeled versus measured concentrations in Cabin John Creek showed some discrepancies.

For the current analysis, we chose not to attempt further calibration of WMOST to better match modelled with measured concentrations because the goal of the current exercise was meeting load reduction targets for Bay TMDLs, not non-tidal water quality targets for criteria. In addition, most observed concentrations for TN, TP, and TSS for the calibration year of 2006 were from base flow periods and did not provide a wide range of concentrations or loads needed for a good calibration. Thus, we used modelled loads based on Phase 6 CBWM time series, and only adjusted CJC TSS time series in WMOST to match estimated loads from the non-tidal Cabin John Creek sediment TMDL. This would have corrected for failure of the CBWM to fully account for inputs from streambank erosion. (Note that continued improvements have been made to the CBWM between the beta 3 version we used and the final version; however, improvements to the

¹ftp://ftp.chesapeakebay.net/Modeling/Phase6/Phase_6_201710/Watershed_Model/WSM_Outputs/Calibration_Figures/04_TOTN_IMAGES_15NOV2017_1732/PL0_4510_0001_TOTN_WINDOW_CONC_15NOV2017_1746.PS.pdf

²ftp://ftp.chesapeakebay.net/Modeling/Phase6/Phase_6_201710/Watershed_Model/WSM_Outputs/Calibration_Figures/05_TOTP_IMAGES_15NOV2017_2350/PL0_4510_0001_TOTP_WINDOW_CONC_16NOV2017_0005.PS.pdf

³ftp://ftp.chesapeakebay.net/Modeling/Phase6/Phase_6_201710/Watershed_Model/WSM_Outputs/Calibration_Figures/06_TSSX_IMAGES_16NOV2017_1109/PL0_4510_0001_TSSX_WINDOW_CONC_16NOV2017_1116.PS.pdf

final Phase 6 model to fully account for all sediment sources may still be needed (Easton et al. 2017).) We did conduct a parallel set of analyses after applying adjustment factors of 1.5 to TN and 0.8 to TP to improve modeled vs observed concentrations. In general, results were qualitatively similar, although greater levels of BMP implementation were required to meet TN load targets, including both dry pond retrofits and the creation of new infiltration basins.

Relative Cost of BMPs Per Unit Load Reduction

Costs per unit volume or pollutant load reduction were calculated for one representative HRU (HRU1 = Montgomery County Turfgrass + Impervious on A+B soils) using BMP costs per impervious acre adjusted for percent impervious area and differences between annual baseline and WMOST calculated BMP-managed loads (Table 14). Although TN, TP, and TSS load reductions calculated by WMOST via SUSTAIN were highest for sand filters, infiltration basins and porous pavement, lower costs associated with dry to wet pond conversions yielded greater cost-efficiencies for this practice. The secondary BMP selected to meet 21% TSS load reductions, infiltration basins, was slightly less cost-effective than sand filters when evaluated on an annual basis, suggesting that infiltration basins were more effective in reducing maximum daily loads. However, sand filters were added to the most cost-effective suite of practices to meet peak flow reduction requirements, consistent with their greater cost effectiveness for annual runoff volume reductions. Porous pavement was the next most cost-effective option for reducing annual peak flows and was also added to the list of practices required for meeting maximum peak flow reductions. Extended dry detention basins were ineffective (with greatest cost/unit reduction) at reducing either annual runoff volume or pollutant loads and were never selected as part of optimization strategies.

Resolution of Differences Among Solutions by Parameter

Optimum load reduction strategies were significantly different for non-tidal TSS TMDL targets as compared to TN and TP targets for the Bay TMDL. TSS solutions were more stringent than Bay TMDL target solutions, not because BMPs are less effective for TSS (Table 14) but because required percentage load reductions for TSS are much higher. If a flow-based strategy is followed, even greater implementation is required to account for sediment inputs from stream bank erosion. However, implementation of solutions for TSS based directly on loads or based on maximum possible peak flow reductions would be more than sufficient for meeting the Bay load reduction targets for TN and TP.

Robustness of Solutions Across Weather Regimes

Optimizations of TSS load reductions (both load and flow-based) were more sensitive to differences in inputs between the average/more variable precipitation year of 2014 and the wet/less variable precipitation year of 2003 than optimizations of TN or TP load reductions. This can be attributed to the fact that modeling of riparian buffer effectiveness in WMOST includes application of a constant percent reduction efficiency

Table 14. Estimated load or volume reductions from HRU1 (Montgomery County turfgrass + impervious on A+B soils) and cost per unit reduction, with capital costs annualized over 10-year period at 5% interest rate. Lowest unit cost/year and pollutant is highlighted in boldface. Mg = million gallons, lb = pounds, mt = metric ton.

BMP Unit:	2014				2003			
	Runoff Vol mg	TN lb	TP lb	TSS mt	Runoff Vol mgd	TN lb	TP lb	TSS mt
2.75" Bioretention Basin	6%	57%	69%	93%	4%	58%	68%	89%
	\$8,746	\$1,920	\$17,419	\$12	\$9,248	\$937	\$11,066	\$9
2.75" Sand Filter w/UD	98%	99%	98%	99%	97%	99%	98%	98%
	\$210	\$401	\$4,466	\$4	\$127	\$198	\$2,770	\$3
2.75" Biofiltration w/UD	4%	46%	59%	89%	2%	45%	56%	84%
	\$25,217	\$3,825	\$33,130	\$20	\$28,572	\$1,928	\$21,624	\$15
2.75" Infiltration Basin	99%	100%	100%	100%	100%	100%	100%	100%
	\$211	\$404	\$4,482	\$4	\$126	\$200	\$2,772	\$3
2.75" Porous Pavement w/UD	100%	100%	100%	100%	100%	100%	100%	100%
	\$1,432	\$2,761	\$30,615	\$27	\$859	\$1,367	\$18,949	\$19
2.75" Extended Dry Detention Basin	-1%	0%	0%	2%	-1%	0%	0%	2%
	---	\$218,910	\$2,989,872	\$15,985,423	---	\$113,678	\$2,051,471	\$305
2.75" Wet Pond Conversion	6%	68%	68%	89%	3%	70%	67%	85%
	\$897	\$162	\$1,792	\$1	\$977	\$78	\$1,118	\$1

regardless of precipitation regime. Most differences in solutions suggested greater implementation would be required to meet goals under 2014 inputs due to constraints imposed by the one large event.

Ancillary Benefits to Flow Regime

Optimization for TSS load reductions has ancillary benefits for low flow and peak flow conditions. Figure 19 shows the WMOST estimate of “baseflow”, the transfer between groundwater and surface water (DQgsw) for baseline conditions, 21% TSS reductions, peak flow reductions, and 100% forested conditions (pre-development). The 21% TSS reduction solution increases baseflow somewhat (18-23%) during the May – September growing season over the baseline conditions for both 2014 and 2003 starting conditions. The flow-optimized solutions increase baseflow much more (456 - 517%) during the growing season, and to a lesser extent during fall and winter months. Effects are more prolonged for the year with above average annual precipitation. Based on 2014 starting conditions, predicted increases actually surpass expected base flows from a simulated 100% forested watershed during fall and winter months.

Transferability of CJC Results to Other MD Watersheds

The CJC watershed is highly developed with little agriculture, complete sewerage, negligible point sources, and no water supply issues. Remaining forest land tends to be concentrated along riparian zones of the main stem of the creek. For similar watersheds in the Piedmont region of Maryland that also have non-tidal sediment TMDLs, we expect the proposed solutions based on WMOST analyses for the CJC to be qualitatively similar. Highly urbanized Maryland watersheds with sediment TMDLs, minimal agriculture, similar percent effective impervious areas, and no WWTP point sources include Bynum Run, Gwynns Falls, and Potomac River. Other similar watersheds but with one or more WWTP point sources include Jones Falls, Rock Creek, and Anacostia River (Appendix H, Figure 20). Watersheds with significant point sources could also have issues with effluent-dominated low flows; these could be mitigated with solutions that increase infiltration and maintain base flows to dilute effluents. However, watersheds with high residual dissolved N concentrations in groundwater from historic agricultural practices may also require implementation of BMPs that treat baseflow and promote denitrification to a greater extent (Hopkins et al. 2017). Watersheds with less intact riparian buffers than the CJC may require greater riparian buffer restoration, particularly if stream temperatures are elevated.

Sources of Uncertainty, Quality and Use of Data

There are several sources of uncertainty related to the analyses presented here, including deficiencies in available calibration and validation data specific to the CJC watershed, uncertainties in urban soil attributes, transferability of BMP performance parameters from New England to Maryland, adjustment of CBWM inputs for hydrologic soil group, performance of riparian zones in urban settings, applicability of O&M costs in SUSTAIN

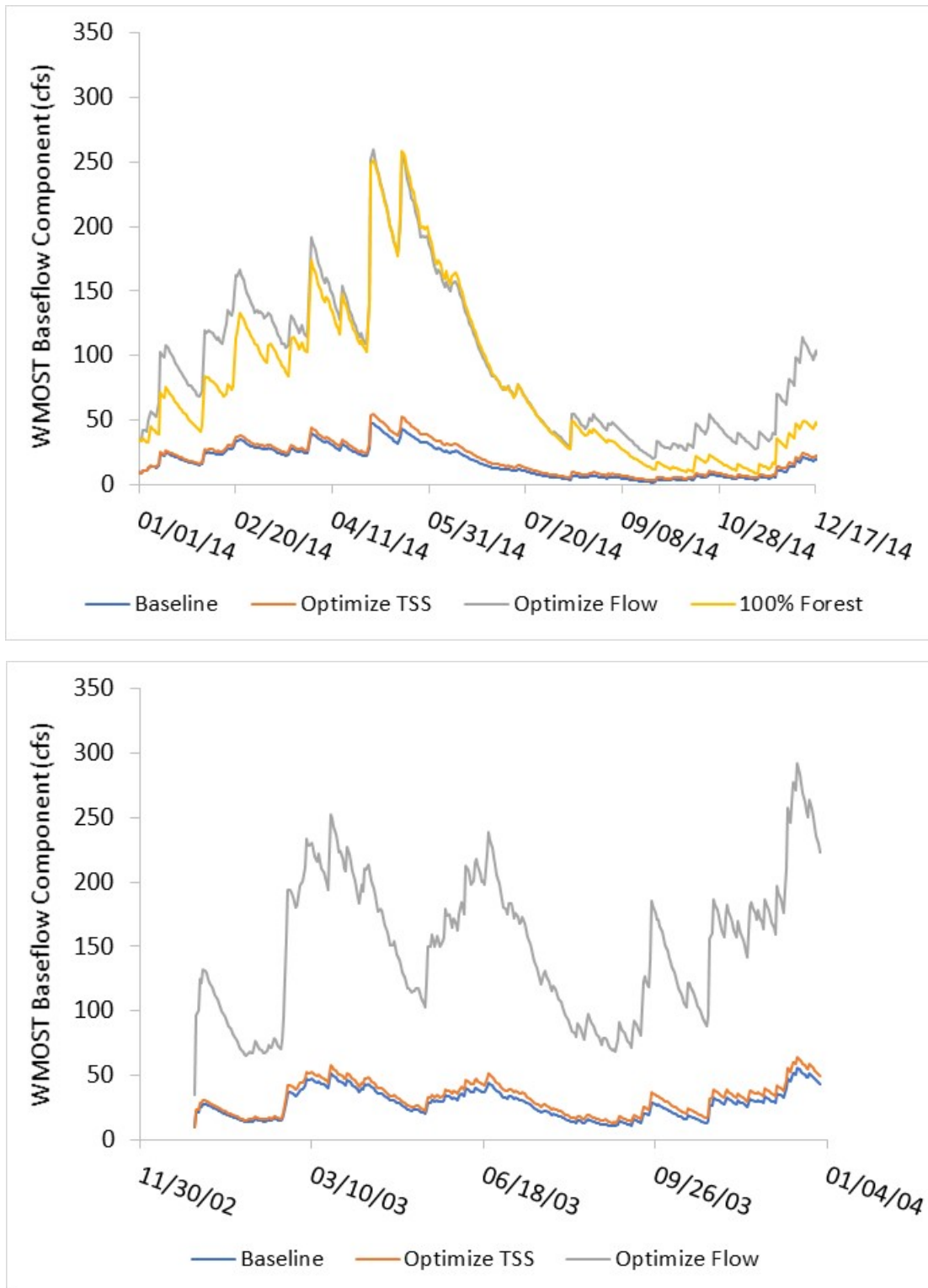


Figure 19. WMOST baseflow component (cfs) for baseline conditions, TSS load reduction optimization, flow reduction optimization, and pre-development (100% forested) conditions for a) 2014 and b) 2003 starting conditions.

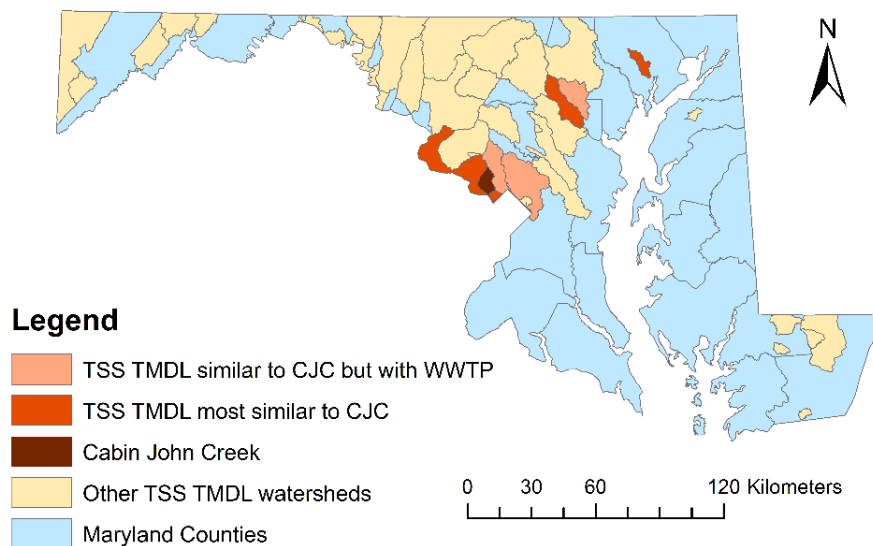


Figure 20. Maryland watersheds with Total Maximum Daily Loads for total suspended solids (TSS) shaded to indicate similarity to Cabin John Creek watershed. See “Transferability to other MD watersheds” for details.

to the CJC watershed, contribution of bank erosion in CJC to TSS loads, and robustness of solutions under future climate change.

Infiltration rates for combined soil hydrologic groups A + B and C + D were assigned based on the midrange of values reported in the CBWM Model documentation (US EPA 2010). However, actual infiltration rates for urban soil/fill material may be lower due to compaction experienced during construction activity (Schwartz and Smith 2016); this would provide overestimates of performance for infiltration-based BMPs.

In this project, we used the default first order decay coefficients provided in WMOST to describe biogeochemical processes taking place within stormwater control measures. These coefficients had been calibrated using actual monitoring data from instrumented stormwater BMPs collected by the University of New Hampshire Stormwater Center but have not been adjusted for differences in temperature regime between New Hampshire and Maryland. Adjustment of coefficients for temperature effects on denitrification would provide increased estimates of BMP performance.

Estimates of riparian buffer performance used in this analysis were based on expert judgement and values from the literature (summarized in Claggett and TetraTech, 2014) and do not reflect potential differences in performance between undeveloped, agricultural, and urban settings. In urban settings, storm sewer inputs can bypass riparian zones, potentially limiting their effectiveness in reducing loadings (trapping particulates and processing interflow) from urban runoff (Roy et al. 2005). However, riparian zones have been demonstrated to reduce nutrient concentrations during baseflow conditions even in suburban settings (Stewart et al. 2006), presumably through enhanced denitrification of groundwater inputs enriched from septic, sewer leakages, and residual

inputs from historic agricultural practices. In nontidal floodplains of the Bay watershed, denitrification potential increases rapidly with percent watershed urbanization up to 10% urban land cover (Korol et al. 2019). The efficiency of denitrification in urban/suburban settings can be reduced by stream downcutting and disconnection from the floodplain (Groffman et al 2003). For floodplains that remain connected with streams in developed watersheds, the stabilizing influence of intact riparian vegetation should still have some effect on bank erosion and sediment trapping during flooding events and should also enhance denitrification by reducing carbon-limitation of denitrification in floodplains (Korol et al. 2019). Sediment budgets constructed for suburban watersheds in the Piedmont have demonstrated that riparian floodplains are still capturing particulate N, P, and sediment in suburban watersheds in the Piedmont (Hopkins et al. 2018) but the net flux (upland + bank sources – bank erosion) varies with stream order and constituent. Riparian zones are still acting as net sinks for particulate N and P deposition, but bank erosion is exceeding floodplain deposition of fine sediments. Bank erosion is greatest from headwater streams (Hopkins et al. 2018).

Operation and maintenance costs can vary significantly among BMP types, municipalities, regions and local settings (e.g., existing development density, accessibility) (ASCE 2017). Although we used BMP construction and design costs specific to Montgomery County, the City of Rockville, and MD State Highway Administration, these local agencies could not provide operation and maintenance (O&M) costs so we relied on default values from SUSTAIN. Table 15 provides a comparison of BMP O&M costs summarized for the state of Maryland with SUSTAIN estimates (King and Hagan 2011). Estimates of O&M costs for Maryland counties adjusted for Montgomery County (factor = 0.985) and updated to \$2016 using the consumer price index (factor of 1.07) are up to an order of magnitude lower than default estimates used by SUSTAIN. However, variation in O&M costs across BMPs is much lower than variation in design and construction costs, so it is unlikely that solutions would differ substantially if generic MD O&M costs were substituted.

Calibration and validation of the WMOST application to measured hydrology and water quality relied heavily on gauging station data from the nearby Anacostia River, a nearby highly urbanized tributary to the Potomac, after adjustment for watershed area. Calibration and validation fit statistics for discharge were satisfactory. Calibration and validation fit statistics for concentrations of TN and TP were based on limited sampling data in CJC, most of which was collected as grab samples during low flow conditions and did not represent the full range of discharge regimes. For the current project, the ability to simulate loadings to meet TMDL loading targets is more critical than the ability to simulate concentrations for water quality criteria. The CBWM, which provided the basis for baseline time series imported to WMOST, has been calibrated with water quality data over a wide range of watershed areas and discharge levels. Overall, the Phase 6 CBWM shows excellent predictions of TN, TP, and TSS loads. The limited instantaneous TN loading data for CJC in 2006 showed a good relationship with WMOST modelled data, with an r^2 of 0.99 (driven by one large value), a low bias at low values (negative intercept) and a slope close to 1. The fit for TSS was not as high ($r^2 = 0.55$), possibly due to challenges in incorporating inputs from stream bank erosion. (Improvements to the

Table 15. Comparison of stormwater BMP O&M costs estimated by WMOST (based on SUSTAIN defaults) as compared to generic MD county-level O&M costs per unit acre of HRU1 (30.87% impervious) based on King and Hagan (2011).

BMP	WMOST O&M cost (\$2016/acre HRU1)	MD O&M cost (\$2016/acre HRU1)
2.75" Bioretention Basin	\$4,765	\$506
2.75" Sand Filter w/UD	\$5,529	\$539
2.75" Infiltration Basin	\$1,923	\$299
2.75" Porous Pavement w/UD	\$1,640	\$1,011
2.75" Extended Dry Detention Basin	\$2,096	\$407
2.75" Wet Pond conversion	\$2,096	\$252

Phase 6 CBWM have substantially increased the estimate of bank erosion inputs as compared to the prior Phase 5 version (https://www.chesapeakebay.net/what/publications/phase_6_modeling_tools). In this case, we chose to adjust the time series input by a factor to ensure that total annual loads matched the edge-of-stream estimate from the non-tidal CJC TMDL. In addition, we provided an alternative approach to estimating necessary load reductions based directly on a sediment-discharge rating curve which should capture bank erosion inputs. Sediment fingerprinting studies in suburban and urban watersheds partially or fully in the Piedmont have shown that the bulk of fine sediments in watershed loads are derived from bank erosion (58 – 91%) as compared to road surfaces (8 – 13%) (Devereux et al. 2010, Cashman et al. 2018).

Comparison with Similar Optimization Studies

Due to the limited number of BMP optimization studies, range of climate regimes and landscape settings, endpoints and combinations of BMPs analyzed, and constraints included in studies, it is difficult to compare our current analyses with previous evaluations. Seventeen studies were identified that had examined effectiveness of different combinations of stormwater control practices including green infrastructure practices in settings ranging from arid to subtropical climates (Appendix I). Only one compared cost-effectiveness of structural vs nonstructural practices (Shoemaker et al. 2013). Most of these studies focused on some form of flow metrics, but a few included water quality endpoints (E coli, phosphorus, or heavy metals). Within analyses focusing on flow metrics, some found more cost-effective solutions for green infrastructure when lower flow reduction targets were implemented, while addition of detention ponds was necessary for peak flow reduction for design storms of greater magnitude (Damodoram et al. 2010). Within assessments focusing on water quality, optimal solutions varied depending on whether event mean concentration or loading targets were evaluated (Baek et al. 2015). Some of these studies only included scenario comparisons rather than full optimization analyses. However, other evaluations went beyond the scope of our analyses, considering optimal size of BMPs, use of site-focused BMPs (rain gardens, rain barrels) versus regional BMPs (detention, bioretention facilities) versus treatment trains, or upstream vs. downstream placement of BMPs in the watershed. None included

comparisons involving both a wide range of stormwater control measures (including infiltration basins) and riparian zones.

Future Improvements

We have only started to address robustness of solutions by comparing BMP performance between years with average and above average precipitation. Initial results suggest that optimization can be influenced more heavily by single large events than by the annual precipitation level. The MDE approach of developing TMDL targets for maximum daily load does take into account temporal variability in daily loads. In the near future we will improve analyses for CJC by further investigating robustness of results under future climate change scenarios.

Conclusions

Optimal solutions differed among constituents considered, the degree of load reductions required, and the relative importance of instream vs watershed sources. Meeting sediment load reduction targets will require greater implementation of green infrastructure stormwater control measures, particularly those with enhanced infiltration. Modelling results show limited effectiveness for gray infrastructure (extended detention basins) in meeting water quality goals.

Although our analysis suggested different least-cost solutions for TN and TP as compared to TSS load reductions, optimal management for the non-tidal sediment TMDL should surpass target reductions for TN and TP for the Bay TMDL. An additional focus on peak flow reduction through infiltration practices would help to limit stream bank erosion, a major contributor to sediment loads in Piedmont streams, and would greatly enhance baseflow. Optimal solutions were relatively robust based on comparisons between 2014 (average precipitation year with one large event) and 2003 (above average precipitation but smaller maximum event). The robustness of solutions was more sensitive to variation in peak flows than to differences in annual precipitation levels.

Unresolved uncertainties include the predicted effectiveness of forested riparian zones in highly urbanized settings in reducing nutrient loads, variation in riparian processing of soluble forms of nutrients in developed landscapes by stream order and flow status (baseflow vs events), and importance of groundwater contributions. More information is needed on the contribution of intact forested riparian zones to bank stabilization and carbon limitation of denitrification processes, and potential interactive effects of riparian zone restoration and upland BMPs.

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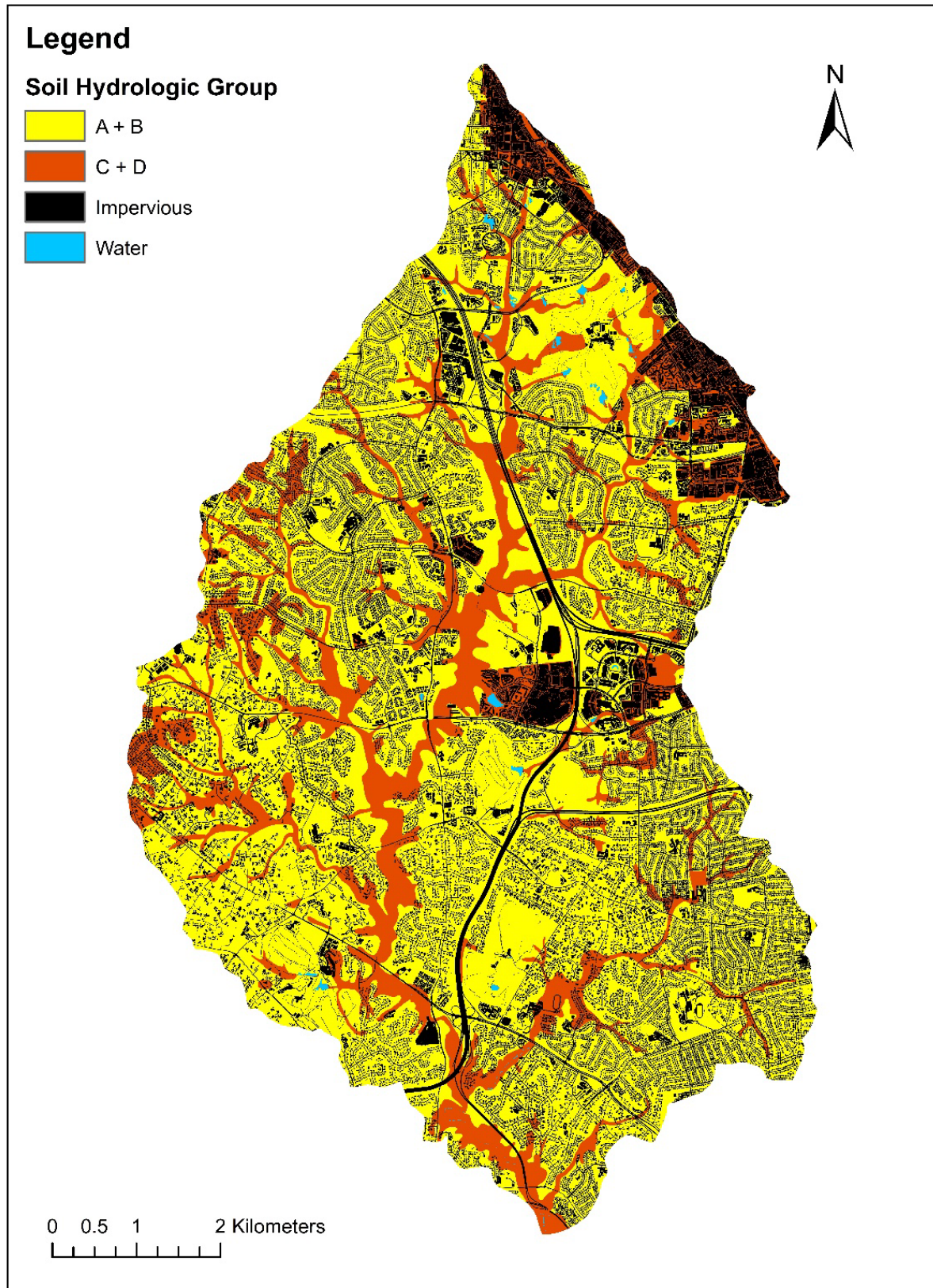
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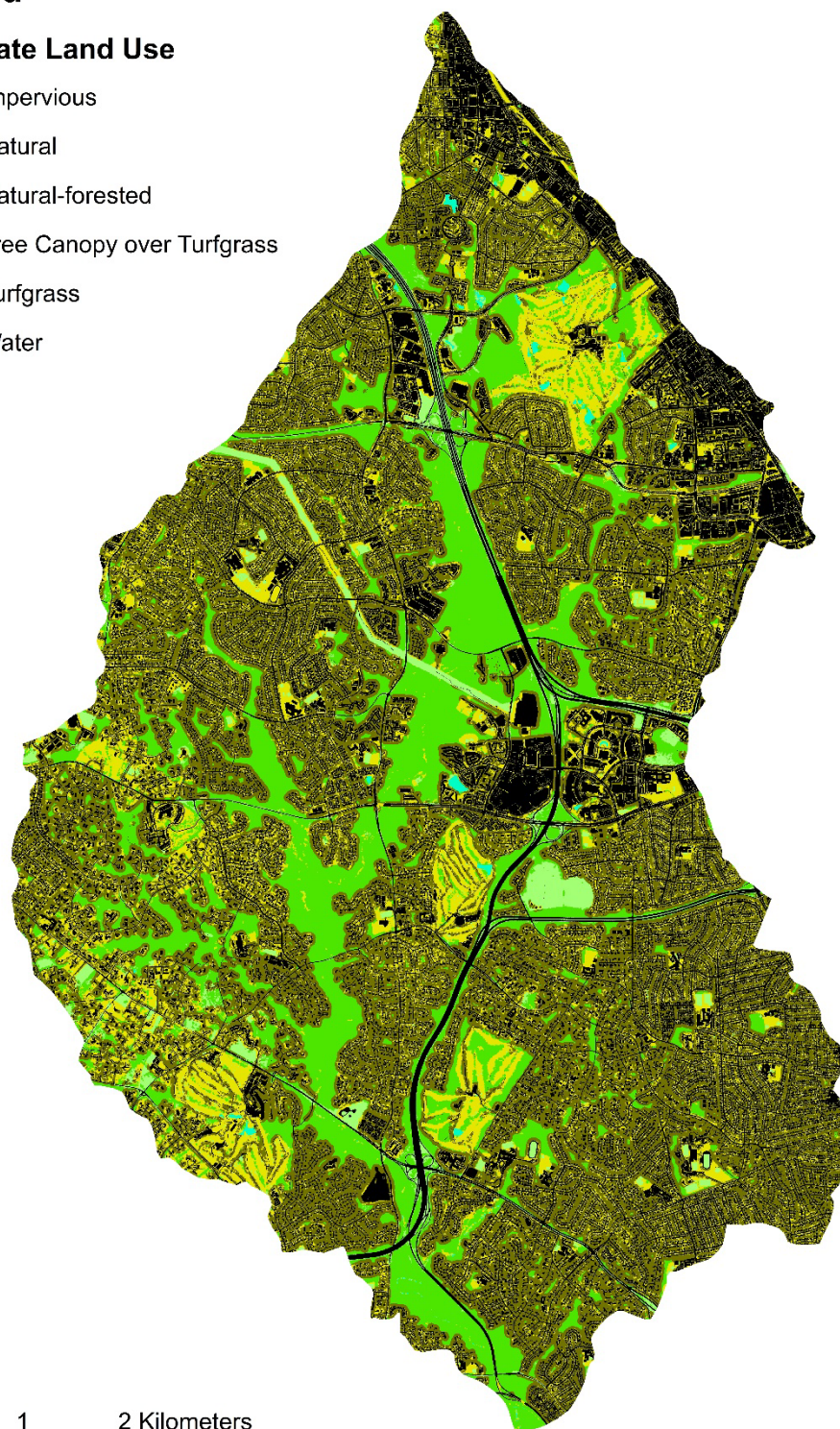
Appendix A. Full-size Maps of HRU Component Distributions



Legend

Aggregate Land Use

- Impervious
- Natural
- Natural-forested
- Tree Canopy over Turfgrass
- Turfgrass
- Water

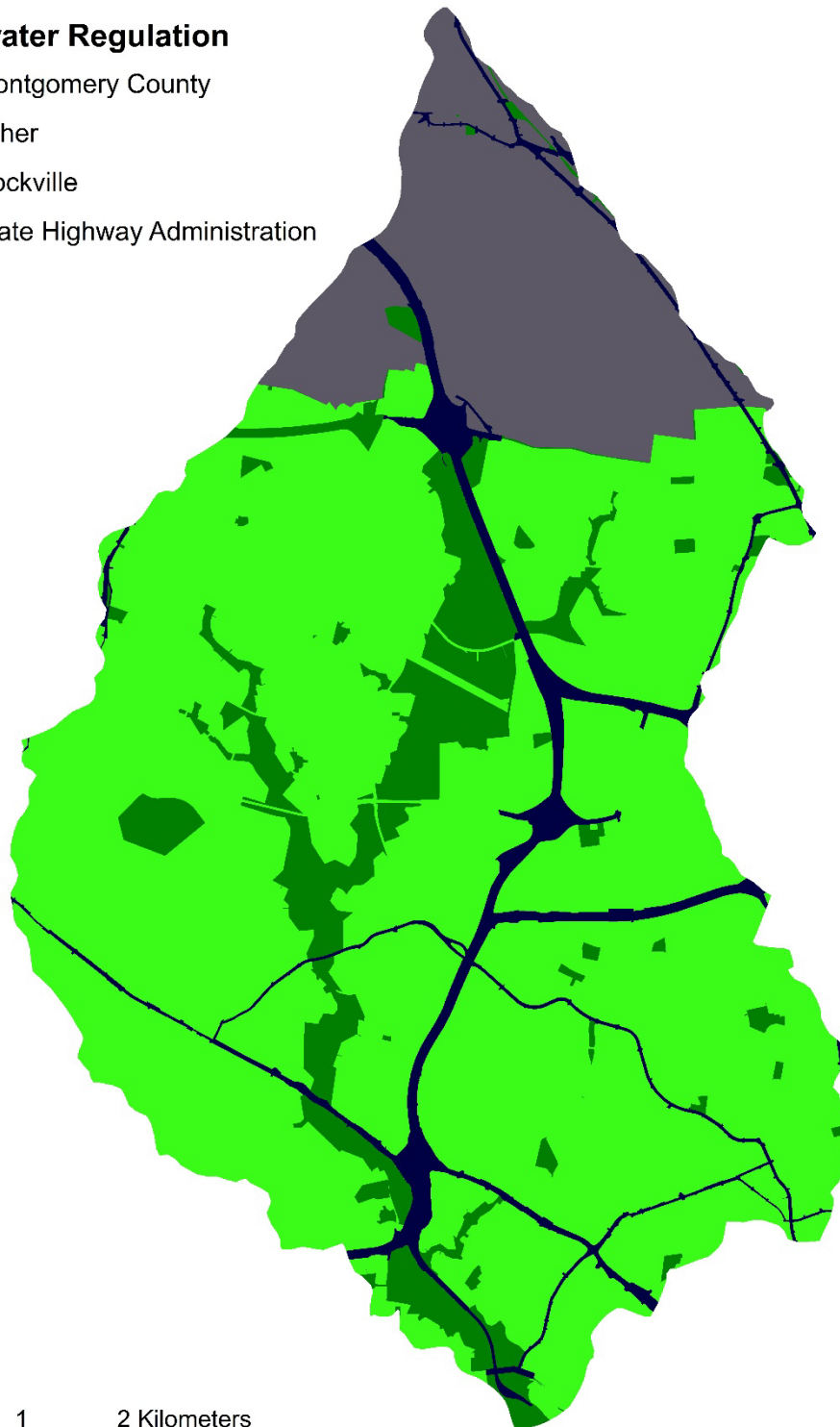


0 0.5 1 2 Kilometers
|-----|-----|-----|-----|

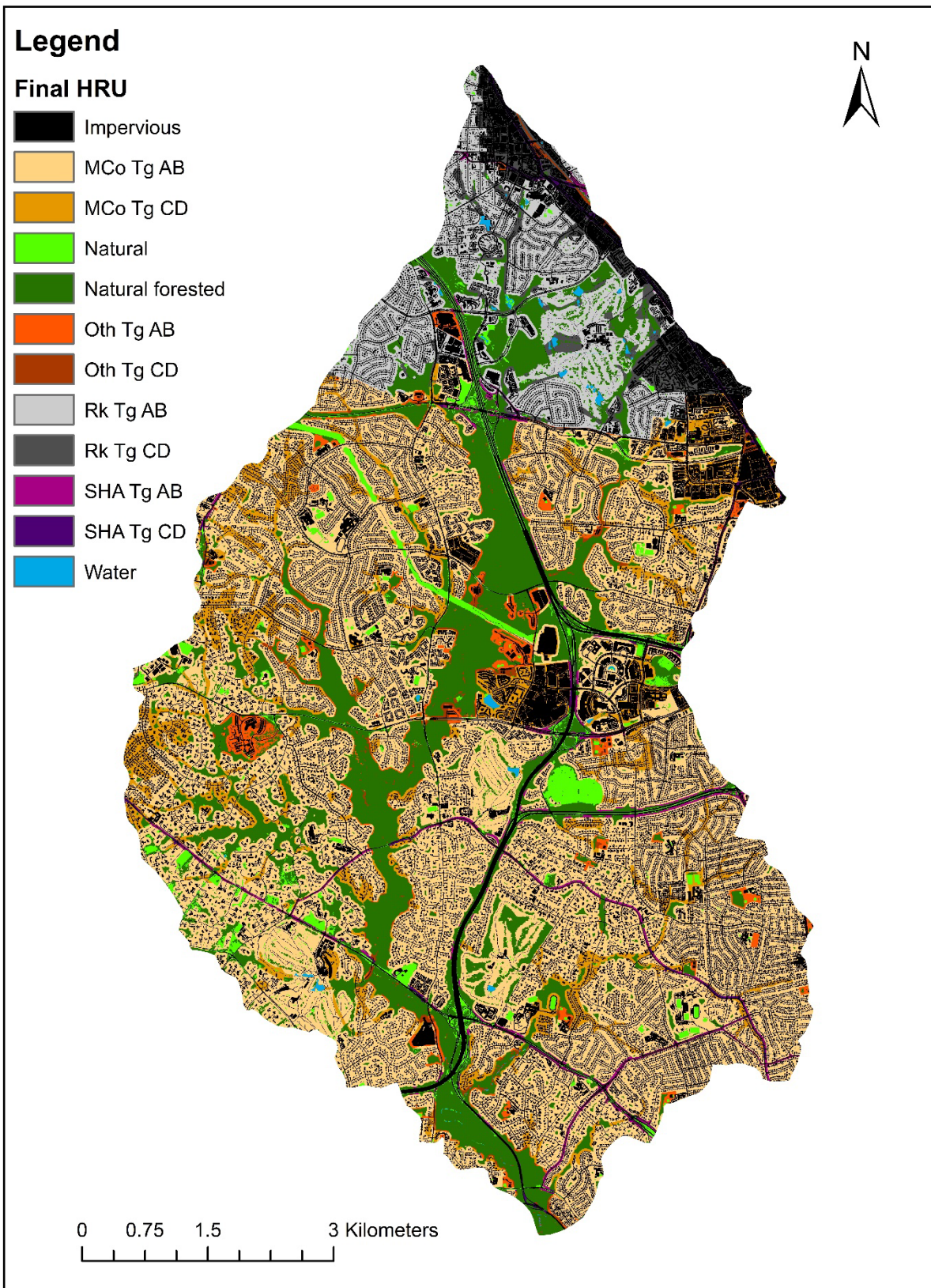
Legend

Stormwater Regulation

- Montgomery County
- Other
- Rockville
- State Highway Administration



0 0.5 1 2 Kilometers



Appendix B: Summary of Data Sources

Appendix C. Phase 6 Land Use Classification

Introduction

There are 39 different land uses in the Phase 6 watershed model (Table C.1) derived from 13 mapped land uses, the County-level Census of Agriculture, and overlays of Municipal Separate Storm Sewer Systems and Combined Sewer Systems. These land uses are further divided into federal and non-federal categories using an overlay of federal lands. In addition, the CBP Partners will use an overlay of wastewater treatment plan service areas to determine population on sewer and septic.

Table C1. Final Phase 6 Land Uses

Developed	Combined Sewer System	Tree Canopy over Turf Grass	cch
		Tree Canopy over Impervious	cci
		Construction	ccn
		Roads	cir
		Buildings and Other	cnr
		Turf Grass	ctg
	Municipal Separate Storm Sewer Systems	Tree Canopy over Turf Grass	mch
		Tree Canopy over Impervious	mci
		Construction	mcn
		Roads	mir
		Buildings and Other	mnr
		Turf Grass	mtg
	Non-regulated Developed Areas	Tree Canopy over Turf Grass	nch
		Tree Canopy over Impervious	nci
		Roads	nir
		Buildings and Other	nnr
		Turf Grass	ntg
Natural		True Forest	for
		Harvested Forest	hfr
		Headwater/isolated Wetland	wto
		Non-tidal Floodplain Wetland	wtf
		Mixed Open	osp
		Water	wat

Agriculture	Commodity Crops	Full Season Soybeans	soy
		Grain without Manure	gom
		Grain with Manure	gwm
		Silage with Manure	swm
		Silage without Manure	som
		Small Grains and Grains	sgg
		Small Grains and Soybeans	sgs
		Other Agronomic Crops	oac
	Hay and forage	Pasture	pas
		Legume Hay	lhy
		Other Hay	ohy
	Specialty Crops	Specialty Crop High	sch
		Specialty Crop Low	scl
	Other	Ag Open Space	aop
		Non-Permitted Feeding Space	fnp
		Permitted Feeding Space	fsp

Mapped Phase 6 Watershed Model Land Uses (with the exception of tidal wetlands)

1. Impervious Roads (IR) = paved and unpaved roads and bridges.
2. Impervious Non-Roads (INR) = buildings, driveways, sidewalks, parking lots, runways, some private roads, railroads and rail right-of-ways, and barren lands within industrial, transitional (early stages of construction), and warehousing land uses.
3. Forest (FOR) = trees farther than 30'-80' from non-road impervious surfaces and forming contiguous patches greater than 1-acre in extent. The variable distance is a result of filtering algorithms (e.g., focal moving windows) applied to the high-resolution non-road impervious surface class.
4. Tree Canopy over Impervious Surfaces (TCI) = trees over roads and non-road impervious surfaces.
5. Tree Canopy over Turf Grass (TCT) = trees within 30'-80' of non-road impervious surfaces where the understory is assumed to be turf grass or otherwise altered through compaction, removal of surface organic material, and/or fertilization.
6. Water (WAT) = all streams, ponds, swimming pools, canals, ditches, wet detention basins, reservoirs, etc. mapped from the high-resolution imagery, NWI ponds & lakes, and synthetic streams derived from a 10m-resolution National

Elevation Dataset using at similar density to those mapped in the 1:24,000-scale National Hydrography Dataset.

7. Tidal Wetlands (WTT) = National Wetlands Inventory (NWI) non-pond, non-lake wetlands, state designated wetlands, and state identified potential wetlands divided into tidal, floodplain, and headwater subclasses. Tidal wetlands include both saline wetlands (E2EM, ESFO, W2SS) and palustrine wetlands (PEM, PFO, PSS) located in a freshwater tidal water regime that was flooded temporally, seasonally, semi-permanently, or permanently. Tidal Wetlands will be included in the water quality model and excluded from the watershed model.
8. Floodplain Wetlands (WTF) = National Wetlands Inventory (NWI) non-pond, non-lake wetlands, state designated wetlands, and state identified potential non-tidal wetlands located within the FEMA designated 100-year floodplain or on frequently flooded soils.
9. Other Wetlands (WTO) = National Wetlands Inventory (NWI) non-pond, non-lake wetlands, state designated wetlands, and state identified potential non-tidal, non-floodplain wetlands. These are typically headwater wetlands or isolated wetlands.
10. Turf Grass (TG) = Herbaceous and barren lands that have been altered through compaction, removal of organic material, and/or fertilization. These include all herbaceous and barren lands within road right-of-ways and residential, commercial, recreational, and other turf-dominated land uses (e.g., cemeteries, shopping centers) and a portion of herbaceous and barren lands within federal facilities, parks, institutional campuses, and large developed parcels.
11. Mixed Open (MO) = All scrub-shrub and herbaceous and barren lands that have been minimally disturbed (e.g., periodically bush hogged, meadows, etc.), reclaimed, or that have internal and/or regulated drainage. These include active, abandoned and reclaimed mines, landfills, beaches, waterbody margins, natural grasslands, utility right-of-ways and a portion of herbaceous lands within industrial, transitional (early stages of construction), and warehousing land uses. Also included are potential agricultural lands that were not mapped as either cropland or pasture in the NASS Cropland Data Layers (2008 through 2015).
12. Cropland (CRP) = Herbaceous and barren lands that are not classed as turf grass or mixed open. The portion of such lands that are crops is determined by the frequency at which the lands are classified as crops in the NASS Cropland Data Layers (2008 through 2015).
13. Pasture/Hay (PAS) = Herbaceous and barren lands that are not classed as turf grass or mixed open. The portion of such lands that are pasture/hay is determined by the frequency at which the lands are classified as pasture/hay in the NASS Cropland Data Layers (2008 through 2015).

Overlays

1. Federal facilities (FED) – all federally owned/managed properties
2. Municipal Separate Storm Sewer Systems (MS4) – areas with Phase I or Phase II stormwater permit. These areas typically drain into municipally owned/operated storm sewer drainage networks within the 2010 Census Urban Areas.
3. Combined Sewer Service Areas (CSS) – areas served by centralized combined wastewater/stormwater treatment systems.
4. Wastewater Treatment Plant Service Areas (SWR) – areas served by public or private sewer utilities.

Appendix D. Selected Screenshots from WMOST Calibration and Optimization Runs.

Screenshot D.1. Land Use tab in WMOST for CJC calibration run. HRU area is set to be equal to baseline areas for land conservation options (min = max) and associated costs are set to -9 to prevent WMOST from implementing any changes in HRU area to implement conservation.

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A1 Land Use and Its Management

Land Use and Its Management

Return to Input Return to Baseline Hydrology

For management options that are not applicable or desired for an HRU, enter -9 for costs.
For Minimum and Maximum Areas, enter -9 if there is no limit on the area for a type of HRU.
Allocation of area among Managed HRU sets is mutually exclusive (i.e., one acre may only receive one management approach/BMP type).
O&M = Operations and maintenance
¹ Data are automatically populated if you use the Hydrology Module.

Baseline HRU Characteristics

HRU ID	HRU Name ¹	Baseline Area [acre]	Percent Effective Impervious ¹	Infiltration Rate [(in/hr) ¹
HRU1B	Turfgrass A/B Montgomery County	8,580	30.9%	0.4
HRU2B	Turfgrass C/D Montgomery County	1,385	30.9%	0.05
HRU3B	Turfgrass A/B City of Rockville	1,460	36.8%	0.4
HRU4B	Turfgrass C/D City of Rockville	352	36.8%	0.05
HRU5B	Turfgrass A/B MD State Highway Administration	428	68.3%	0.4
HRU6B	Turfgrass C/D MD State Highway Administration	125	68.3%	0.05
HRU7B	Turfgrass A/B Other Regulated	209	22.7%	0.4
HRU8B	Turfgrass C/D Other Regulated	61	22.7%	0.05
HRU9B	Natural	3,637	0.0%	0.12308
HRU10B	Water	35	0.0%	0

Data for Land Conservation Option

Minimum Area [acre]	Maximum Area [acre]	Initial Cost to Conserve [\$ /acre]	O&M Cost [\$ /acre/yr]
8580	8580	-9	-9
1385	1385	-9	-9
1460	1460	-9	-9
352	352	-9	-9
428	428	-9	-9
125	125	-9	-9
209	209	-9	-9
61	61	-9	-9
3637	3637	-9	-9
35	35	-9	-9

Ready

Screenshot D.2. Potable demand tab in WMOST calibration run. All drinking water demand in the CJC watershed is met through interbasin transfers so the number of water user classes was set to 1 on the Input Data Screen (not shown) and each element of the unaccounted water (UAW) and user demand and consumption time series was set to zero.

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Potable Demand

Return to Input

Enter data in blue input fields for available time period. Time series must be consecutive and complete, e.g., no skipped days.
For monthly time step, the day of the month does not matter. Enter data in blue input fields for each water use for each month.

Date (mm/dd/yyyy)	Total Water Demand [million gallons / time step]	Unaccounted	residential	Average Percent Consumptive Water Use (%)
1/1/2006	0.00	0.00	0.00	January 0
1/2/2006	0.00	0.00	0.00	February 0
1/3/2006	0.00	0.00	0.00	March 0
1/4/2006	0.00	0.00	0.00	April 0
1/5/2006	0.00	0.00	0.00	May 0
1/6/2006	0.00	0.00	0.00	June 0
1/7/2006	0.00	0.00	0.00	July 0
1/8/2006	0.00	0.00	0.00	August 0
1/9/2006	0.00	0.00	0.00	September 0
1/10/2006	0.00	0.00	0.00	October 0
1/11/2006	0.00	0.00	0.00	November 0
1/12/2006	0.00	0.00	0.00	December 0
1/13/2006	0.00	0.00	0.00	
1/14/2006	0.00	0.00	0.00	
1/15/2006	0.00	0.00	0.00	
1/16/2006	0.00	0.00	0.00	
1/17/2006	0.00	0.00	0.00	
1/18/2006	0.00	0.00	0.00	
1/19/2006	0.00	0.00	0.00	
1/20/2006	0.00	0.00	0.00	
1/21/2006	0.00	0.00	0.00	
1/22/2006	0.00	0.00	0.00	
1/23/2006	0.00	0.00	0.00	
1/24/2006	0.00	0.00	0.00	
1/25/2006	0.00	0.00	0.00	

Enter average daily wastewater loadings of each constituent from each potable water user.

Constituent	Loadings (lbs/time step)
TN	residential

Intro Input Potable Demand

Ready 100%

Screenshot D.3. Septic_Sewer tab in WMOST calibration run. Virtually all CJC watershed is sewered so percentage public water users on septic was set to 0%.

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A1 Septic System Users and Sewer Use

Septic System Users and Sewer Use

Enter the daily average effluent concentration for septic treatment of each constituent.

Customers with Public Water & Septic Systems Recharge	Constituent	Concentration (mg/L)
residential	TN	0

0

Enter the daily average effluent concentration for enhanced septic treatment of each constituent.

Customers with Public Water & Septic Systems Recharge	Constituent	Concentration (mg/L)
residential	TN	0

0

Note: Some user types discharge to the wastewater treatment rows need to add to 100%. Each value is the percent septic use for a user type.

Enhanced Septic Treatment

O&M cost for enhanced septic capacity	\$/MG
Maximum capacity of enhanced septic	MGD

Storm Sewer

Maximum capacity of storm sewer	MGD	-9
---------------------------------	-----	----

To exclude limits on the storm sewer, enter -9.

Municipal flows that go directly to sewer system (%)

Month	Municipal
January	0
February	0
March	0
April	0
May	0
June	0
July	0
August	0
September	0
October	0
November	0
December	0

Sanitary Sewer

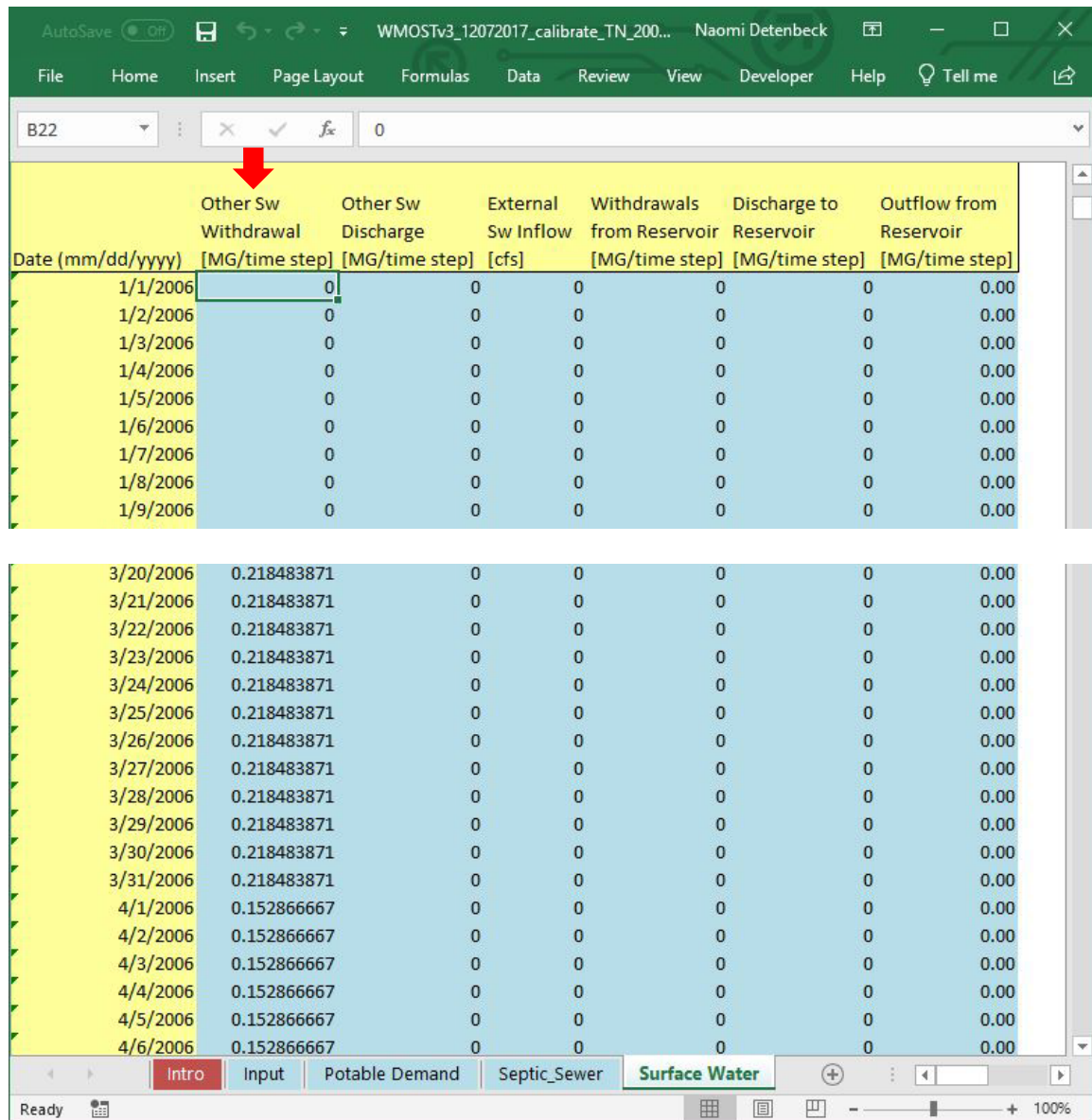
Maximum capacity of sanitary sewer	MGD	-9
------------------------------------	-----	----

To exclude limits on the sanitary sewer, enter -9.

Intro Input Potable Demand **Septic Sewer**

Ready 100%

Screenshot D.4. Surface water tab in WMOST calibration run. Private withdrawals and private discharge time series to surface water and groundwater were entered based on data obtained by MDE from withdrawal and discharge permits.



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B22

Date (mm/dd/yyyy)	Other Sw Withdrawal [MG/time step]	Other Sw Discharge [MG/time step]	External Sw Inflow [cfs]	Withdrawals from Reservoir [MG/time step]	Discharge to Reservoir [MG/time step]	Outflow from Reservoir [MG/time step]
1/1/2006	0	0	0	0	0	0.00
1/2/2006	0	0	0	0	0	0.00
1/3/2006	0	0	0	0	0	0.00
1/4/2006	0	0	0	0	0	0.00
1/5/2006	0	0	0	0	0	0.00
1/6/2006	0	0	0	0	0	0.00
1/7/2006	0	0	0	0	0	0.00
1/8/2006	0	0	0	0	0	0.00
1/9/2006	0	0	0	0	0	0.00
3/20/2006	0.218483871	0	0	0	0	0.00
3/21/2006	0.218483871	0	0	0	0	0.00
3/22/2006	0.218483871	0	0	0	0	0.00
3/23/2006	0.218483871	0	0	0	0	0.00
3/24/2006	0.218483871	0	0	0	0	0.00
3/25/2006	0.218483871	0	0	0	0	0.00
3/26/2006	0.218483871	0	0	0	0	0.00
3/27/2006	0.218483871	0	0	0	0	0.00
3/28/2006	0.218483871	0	0	0	0	0.00
3/29/2006	0.218483871	0	0	0	0	0.00
3/30/2006	0.218483871	0	0	0	0	0.00
3/31/2006	0.218483871	0	0	0	0	0.00
4/1/2006	0.152866667	0	0	0	0	0.00
4/2/2006	0.152866667	0	0	0	0	0.00
4/3/2006	0.152866667	0	0	0	0	0.00
4/4/2006	0.152866667	0	0	0	0	0.00
4/5/2006	0.152866667	0	0	0	0	0.00
4/6/2006	0.152866667	0	0	0	0	0.00

Intro Input Potable Demand Septic Sewer **Surface Water**

Ready 100%

Screenshot D.5. Groundwater tab in WMOST calibration run. Infiltration and inflow to the sewer was set at 2.18 million gallons per day (MGD; City of Rockville Master Plan).

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A1 Groundwater

Groundwater

Return to Input

Groundwater recession coefficient 0.03000 [1/time step]

Initial groundwater volume 200 [MG]

Minimum volume 10 [MG]

Maximum volume 5,000 [MG]

Cost for make-up groundwater 1,000,000 \$/MG

To exclude make-up water penalty, enter -9.

If you used the Baseline Hydrology module, you may automate the calculation of the groundwater recession coefficient.

Calculate and Populate the Groundwater Recession Coefficient

Enter initial water quality and removal rates in the groundwater system.

Groundwater

TN

Initial Condition: Enter the initial constituent concentration in groundwater

Concentration (mg/L) 2

Attenuation Rate: Enter a removal rate value and -9 for the exclusion of the rate

0th order rate (mg/L/time step) -9

OR

1st order rate (1/time step) -9

TN Loadings Time Series

Private Gw Point External Gw

Source Loadings Loadings

[Lbs/time step] [Lbs/time step]

Enter a minimum surface watershed

Month

Date [mm/dd/yyyy]	Other Gw Withdrawal [MG/time step]	Other Gw Discharge [MG/time step]	External Gw Inflow [MG/time step]
1/1/2006	2.18	0.00	0.00
1/2/2006	2.18	0.00	0.00
1/3/2006	2.18	0.00	0.00
1/4/2006	2.18	0.00	0.00
1/5/2006	2.18	0.00	0.00
1/6/2006	2.18	0.00	0.00
1/7/2006	2.18	0.00	0.00
1/8/2006	2.18	0.00	0.00
1/9/2006	2.18	0.00	0.00
1/10/2006	2.18	0.00	0.00
1/11/2006	2.18	0.00	0.00
1/12/2006	2.18	0.00	0.00
1/13/2006	2.18	0.00	0.00
1/14/2006	2.18	0.00	0.00
1/15/2006	2.18	0.00	0.00
1/16/2006	2.18	0.00	0.00
1/17/2006	2.18	0.00	0.00
1/18/2006	2.18	0.00	0.00
1/19/2006	2.18	0.00	0.00
1/20/2006	2.18	0.00	0.00
1/21/2006	2.18	0.00	0.00

Month	Private Gw Point Source Loadings [Lbs/time step]	External Gw Loadings [Lbs/time step]
January	0.00	0.00
February	0.00	0.00
March	0.00	0.00
April	0.00	0.00
May	0.00	0.00
June	0.00	0.00
July	0.00	0.00
August	0.00	0.00
September	0.00	0.00
October	0.00	0.00
November	0.00	0.00
December	0.00	0.00

Ready

Intro Input CSO Potable Demand Septic Sewer Surface Water Groundwater Infrastructure

100%

Screenshot D.6. Infrastructure tab for WMOST calibration run. Infrastructure modification options were set to -9 with associated costs set to zero.

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A1 Infrastructure

Infrastructure
Capacity, cost, and management data for water treatment plant, potable distribution system, wastewater treatment plant, water reuse facility, nonpotable distribution system, and aquifer storage and recharge

Return to Input

Planning horizon [years] 10 Planning horizon is the time period over which capital costs are annualized
Interest rate [%] 5.00 unless the lifetime of the management practice or infrastructure is requested.

Water services

	Value	Units
Consumer's price for potable water: Fixed fee	0	\$/month
Consumer's price for potable water: Variable, volume-based fee	0.00	\$/HCF

To exclude new and additional capacity for a facility, enter -9 in the corresponding blue box.

Groundwater (Gw) Pumping

	Value	Units	Exclude New/Additional Capacity
Capital cost for additional capacity		\$/MGD	-9
Operation & Maintenance (O&M) costs		\$/MG	
Existing maximum capacity	0.00	MGD	
Lifetime remaining on existing infrastructure	0	yrs	
Lifetime of new construction	0	yrs	

Surface Water (Sw) Pumping

	Value	Units	Exclude New/Additional Capacity
Capital cost for additional capacity		\$/MGD	-9
O&M costs		\$/MG	
Existing maximum capacity	0.00	MGD	
Lifetime remaining on existing infrastructure	0	yrs	
Lifetime of new construction	0	yrs	

Water Treatment Plant (WTP)

	Value	Units	Exclude New/Additional Capacity
Capital cost for additional capacity		\$/MGD	-9
O&M costs		\$/MG	
Existing maximum capacity	0.00	MGD	
Lifetime remaining on existing infrastructure	0	yrs	
Lifetime of new construction	0	yrs	

Unaccounted-for-Water/ Potable water distribution system leak

	Value	Units
Initial cost for survey & repair	0	\$
O&M costs for maintaining reduction in UAW	0	\$/yr
Maximum percent UAW that can be fixed		%

HCF = hundred cubic feet
MGD = million gallons per day
MG = million gallons
O&M = operations and maintenance
UAW = Unaccounted for Water

Water Treatment Plant (WTP)

	Value	Units
Maximum daily influent		MGD
Average daily effluent		MGD

Upgraded Water Treatment Plant (WTP)

	Value	Units
Capital cost for additional capacity		\$/MGD
Lifetime of new construction	0	yrs

Upgraded Water Treatment Plant (WTP)

	Value	Units
Capital cost for additional capacity		\$/MGD
Lifetime of new construction	0	yrs

Ready

Screenshot D.7. Riparian buffer BMP module for WMOST optimization run. Riparian pixels in potential restoration areas were ranked by receiving load and then categorized into low, medium, and high relative load groups based on order-of-magnitude differences.

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P15 Riparian Buffer Implementation Mode

Return to Input Return to Stormwater Hydrology Module

Riparian Buffer BMPs

Follow the step-by-step directions to apply riparian buffer land area conversions. Based on your selections, the model may convert the riparian area land use to meet any specified targets.

1. Riparian Buffer Management: Convert riparian buffer land use to reduce or increase riparian area and upland area loadings.

1A. Riparian Buffer Management Conversion Sets

Enter the number of land use conversions: (max 10) **Setup Riparian Buffer Tables**

Enter the number of relative loads groups: (max 5)

1. Select the HRU types that are available for land use conversion from **From HRU Name** and **To HRU Name**. The **From HRU ID** and **To HRU ID** will automatically populate from the baseline land use table.

2. Enter the following data into the tables below:

- 1) riparian area available for land use conversion (acres) based on the relative loads the land area rec
- 2) the capital costs of converting the area (\$/acre),
- 3) the O&M costs of converting the area (\$/acre/yr),
- 4) the load adjustment efficiency that is applied to upgradient areas only, and
- 5) the upland areas associated with each riparian area relative loads group in Table 1B.

Land Use Conversion No.	From HRU ID	To HRU ID	From HRU Name	To HRU Name	Riparian Area, Relative Loads Group 1 (acre)	Riparian Area, Relative Loads Group 2 (acre)	Riparian Area, Relative Loads Group 3 (acre)	Initial Cost to Convert (\$/acre)	O&M Cost (\$/acre/yr)	TN Load Adj Efficiency (%)	Riparian Buffer Implementation Mode
16	RBM1LC1	1	10 Turfgrass A/B Montgomery County	Natural	25.73	107.38	0.04	20	0.22	-25	Optimization
17	RBM1LC2	2	10 Turfgrass C/D Montgomery County	Natural	32.30	102.64	0.12	20	0.22	-25	Optimization
18	RBM1LC3	3	10 Turfgrass A/B City of Rockville	Natural	15.46	17.54	0.50	20	0.22	-25	Optimization
19	RBM1LC4	4	10 Turfgrass C/D City of Rockville	Natural	6.79	15.63	0.65	20	0.22	-25	Optimization
20	RBM1LC5	5	10 Turfgrass A/B MD State Highway Administration	Natural	2.37	2.52	0.00	20	0.22	-25	Optimization
21	RBM1LC6	6	10 Turfgrass C/D MD State Highway Administration	Natural	1.88	3.52	0.00	20	0.22	-25	Optimization
22	RBM1LC7	7	10 Turfgrass A/B Other Regulated	Natural	1.43	12.72	0.06	20	3.599	-25	Optimization
23	RBM1LC8	8	10 Turfgrass C/D Other Regulated	Natural	1.78	19.77	1.17	20	3.599	-25	Optimization
24	RBM1LC9	9	10 Natural-nonforested	Natural	1.86	17.01	1.51	20	0.22	-25	Optimization

Note: a negative load adjustment value represents a reduction in loadings, while a positive load adjustment value represents an increase in loadings.

1B. Riparian Buffer Management Upland Land Use

HRU ID	HRU Name	Upland Area, Relative Loads Group 1 (acres)	Upland Area, Relative Loads Group 2 (acres)	Upland Area, Relative Loads Group 3 (acres)
31	RBM1HRU1 Turfgrass A/B Montgomery County	991.18	1,990.48	1.28
32	RBM1HRU2 Turfgrass C/D Montgomery County	107.77	300.05	0.07
33	RBM1HRU3 Turfgrass A/B City of Rockville	328.55	123.37	0.82
34	RBM1HRU4 Turfgrass C/D City of Rockville	51.78	39.30	0.14
35	RBM1HRU5 Turfgrass A/B MD State Highway Administration	28.08	39.10	0.00
36	RBM1HRU6 Turfgrass C/D MD State Highway Administration	9.64	8.23	0.00
37	RBM1HRU7 Turfgrass A/B Other Regulated	31.78	76.86	6.27
38	RBM1HRU8 Turfgrass C/D Other Regulated	8.38	22.04	1.05
39	RBM1HRU9 Natural-nonforested	44.46	129.03	3.69
40	RBM1HRU10 Natural			
41	RBM1HRU11 Water			

Intro Input Riparian Buffers

Ready 85%

Screenshot D.8. Land use tab from WMOST optimization run. We subtracted areas treated by existing BMPs in 2014 from total suitable HRU areas to obtain maximum HRU area available for treatment.

AutoSave (On) WMOSTv3.01_09282018_optimize_TN_2014_noadj_7BMPs_gwrpt03.xlsm - Naomi Detenbeck

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A1 Land Use and Its Management

	A	B	C	D	E	F	G	H
60	First Set of Managed Land Uses and Their Limits					2.75" Bioretention Basin	< Input name of management practice	
61	HRU ID	HRU Name	Minimum Area [acre]	Maximum Area [acre]	Initial Cost to Conserve [\$/acre]	O&M Cost [\$/acre/yr]		
62	HRU1M1	Turfgrass A/B Montgomery County	0	9039.852282	168.8	4.764613537		
63	HRU2M1	Turfgrass C/D Montgomery County	0	1459.363887	168.8	4.764613537		
64	HRU3M1	Turfgrass A/B City of Rockville	0	1543.954752	382.319279	5.684277173		
65	HRU4M1	Turfgrass C/D City of Rockville	0	371.7947004	382.319279	5.684277173		
66	HRU5M1	Turfgrass A/B MD State Highway Administration	0	444.7499534	211.195216	10.54770447		
67	HRU6M1	Turfgrass C/D MD State Highway Administration	0	129.9628793	211.195216	10.54770447		
68	HRU7M1	Turfgrass A/B Other Regulated	0	332.7820299	260.489834	3.500094808		
69	HRU8M1	Turfgrass C/D Other Regulated	0	121.8529445	260.489834	3.500094808		
70	HRU9M1	Natural	0	0	-9	-9		
71	HRU10M1	Water	0	0	-9	-9		
112	Second Set of Managed Land Uses and Their Limits					2.75" Sand Filter w/UD	< Input name of management practice	
114	HRU ID	HRU Name	Minimum Area [acre]	Maximum Area [acre]	Initial Cost to Conserve [\$/acre]	O&M Cost [\$/acre/yr]		
115	HRU1M2	Turfgrass A/B Montgomery County	0	8894.416254	52.5	5.528924118		
116	HRU2M2	Turfgrass C/D Montgomery County	0	1436.741212	52.5	5.528924118		
117	HRU3M2	Turfgrass A/B City of Rockville	0	1517.889666	207.737802	6.59611465		
118	HRU4M2	Turfgrass C/D City of Rockville	0	364.5897929	207.737802	6.59611465		

Intro Input Land Use

Ready 100%

Screenshot D.9. Stormwater Hydrology and Loadings Module from WMOST optimization run. Structural BMPs tested included Bioretention (SUSTAIN = Bioretention Basin), Filtering Practices (SUSTAIN = Sand Filter), Infiltration Practices (SUSTAIN = Infiltration Basin), Permeable Pavement (SUSTAIN = Porous Pavement w/UD), and Dry Detention Ponds (SUSTAIN = Extended Dry Detention Basin).

AutoSave WMOSTv3.01_09282018_optimize_TN_2014_noadj_7BMPs_gwrpt03.xlsx - Excel Naomi Detenbeck

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A1

Return to Input

Stormwater Hydrology and Loadings Module

Follow the step-by-step directions to process baseline hydrology and loadings data and obtain stormwater managed hydrology and loadings. Based on your selections, the model will populate input fields on the appropriate tabs.

To use this Stormwater Module, you MUST enter baseline hydrology and loadings data first. You may do so manually or via the Hydrology Module.

1. Did you use the Baseline Hydrology and Loadings Module or manually enter data? Entered Own Data

If you used the Baseline Hydrology and Loadings Module, use the Import Hourly Time Series button to automatically populate hourly data. Import Hourly Time Series OR If you entered data manually, use the Hourly Time Series button to manually enter hourly data. Manually Enter Hourly Data

2. Model time step Did you run a daily or monthly model? Daily

3. Select the best management practices (BMPs) that you would like to model and enter the design depth for each BMP. Indicate the decay rate of water quality constituents for each BMP. Enter 0 for no constituent decay. Use the Import Default Decay Rates button to use default values. See the User's Guide for descriptions of the BMPs and guidance on sizing for different water management objectives and default BMP design values. Each BMP will be sized and run for all undeveloped HRUs (EIA = 0%).

BMP Type **Design Depth For BMP [in]** **TN 1st Order Decay Rate [1/hr]**

Bioretention Basin	2.75	0.03
Sand Filter w/UD	2.75	0.36
Biofiltration w/UD	2.75	0.03
Infiltration Basin	2.75	0.42
Porous Pavement w/UD	2.75	0.26
Extended Dry Detention Basin	2.75	0.02
Wet Pond	2.75	0.03

Import Default Decay Rates

Area Managed by Agriculture BMP [acres] **TN Settling Velocity [in/s]**

4. Agricultural BMPs If you want to model agricultural BMPs, use the button to the right to show the inputs cells for additional BMP parameter inputs. Show Agricultural BMP Inputs

5. Use the buttons below to perform the indicated processes. If you plan to use any agricultural BMPs, see the additional inputs in Step 6 before creating the SUSTAIN input files.

Intro Input Stormwater Land Use +

Ready

Screenshot D.10. Water Quality BMP module from WMOST optimization run. Potential alternative BMPs tested included streambank stabilization/restoration.

AutoSave WMOSTv3.01_09282018_optimize_TN_2014_noadj_78BMPstrbnkrestn_gwrpt03.xlsx Naomi Detenbeck

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A1

Water Quality BMPs

Return to Input Return to Stormwater Hydrology Module

Follow the step-by-step directions to apply water quality BMPs to the stream, reservoir, or land use. Based on your selections, the model will populate input fields on the appropriate tabs.

1. Loadings Target Adjustment BMPs: Apply a loadings credit to the loadings target in the stream or reservoir.

1A. Streambank Restoration or Stabilization
 Enter the constituent removal rate resulting from a streambank project.

Removal Rate of TN 0.000205479 lb/ft/time step

Enter the cost per stream length treated and the maximum stream length that can be treated.

Cost per linear stream length 0.44710541 \$/foot thous \$
 Maximum linear stream length 214022.3166 feet

Specify whether the stabilization project applies to the in-stream or reservoir loading target. In-Stream Loadings Target

1B. Outfall Enhancement or Stabilization
 Enter the constituent removal rate resulting from an outfall project.

Removal Rate of TN lb/outfall/time step

Enter the cost per outfall treated and the maximum number of outfalls that can be treated.

Cost per outfall treated \$/outfall
 Maximum number of outfalls treated outfalls

Specify whether the outfall enhancement project applies to the in-stream or reservoir loading target.

2. Runoff Loadings Direct Reduction BMPs: Apply direct runoff loadings reduction to an HRU area by implementing a direct reduction BMP.

2A. Runoff Loadings Direct Reduction Management Sets

You can model the following three management options:

Select the name of the direct reduction BMP(s) that you would like to model.

1) Street Sweeping
 2) Tree Canopy over Impervious/Turf Area
 3) Urban Nutrient Management

Select the number of direct reduction managed sets: 0

Setup Direct Reduction Tables

Set No.	Direct Reduction Management Name
DRM1	Street Sweeping
DRM2	
DRM3	

Enter the application area, cost of application, and load adjustment for each constituent in the tables below.
 If an HRU does not receive any treatment from the management practice, enter 0 for the load removal rate.

2B. Runoff Loadings Direct Reduction Management Land Uses:

Ready

Screenshot D.11. Water Quality BMP module for WMOST optimization run. Nonstructural BMPs tested included street sweeping and tree canopy over turf.

Water Quality BMPs

Follow the step-by-step directions to apply water quality BMPs to the stream, reservoir, or land use. Based on your selections, the model will populate input fields on the appropriate tabs.

1. Loadings Target Adjustment BMPs: Apply a loadings credit to the loadings target in the stream or reservoir.

2. Runoff Loadings Direct Reduction BMPs: Apply direct runoff loadings reduction to an HRU area by implementing a direct reduction BMP.

2A. Runoff Loadings Direct Reduction Management Sets

Select the name of the direct reduction BMP(s) that you would like to model:

BMP No.	Direct Reduction Management Name
DRM1	Tree Canopy over Impervious/Turf
DRM2	
DRM3	

You can model the following three management options:

- Street Sweeping
- Tree Canopy over Impervious/Turf Area
- Urban Nutrient Management

Select the number of direct reduction managed sets:

Setup Direct Reduction Tables

Enter the application area, cost of application, and load adjustment for each constituent in the tables below.
If an HRU does not receive any treatment from the management practice, enter 0 for the load removal rate.

2B. Runoff Loadings Direct Reduction Management Land Uses:

First Set of Direct Reduction Application Areas and their Costs

HRU ID	HRU Name	Application Area [acre]	Initial Cost of Application [\$ / acre]	O&M Cost [\$ / acre / yr]	TN Load Removal Rate [%]
HRU1DRM1	Turfgrass A/B Montgomery County	4555.486135	23.01	1.38	23.80
HRU2DRM1	Turfgrass C/D Montgomery County	724.9813413	23.01	1.38	23.80
HRU3DRM1	Turfgrass A/B City of Rockville	980.8894522	23.01	1.38	18.16
HRU4DRM1	Turfgrass C/D City of Rockville	250.8852159	23.01	1.38	18.16
HRU5DRM1	Turfgrass A/B MD State Highway Administration	338.4843239	23.01	1.38	13.34
HRU6DRM1	Turfgrass C/D MD State Highway Administration	106.3402503	23.01	1.38	13.34
HRU7DRM1	Turfgrass A/B Other Regulated	158.3656952	23.01	1.38	20.33
HRU8DRM1	Turfgrass C/D Other Regulated	64.39972174	23.01	1.38	20.33
HRU9DRM1	Natural	0.00	0.00	0.00	0.00
HRU10DRM1	Water	0.00	0.00	0.00	0.00

Appendix E1. Summary of costs of best management practices for structural and nonstructural (alternative) applications by regulated entity in Montgomery County, MD.

Appendix E2. Raw data used in calculating summary of costs of best management practices for structural and nonstructural (alternative) applications by regulated entity in Montgomery County, MD.

Appendix F. 2014 HRU areas and history of implementation of existing stormwater BMPs in Cabin John Creek

Appendix G. MDE Estimates of pollutant removal efficiencies for structural, alternative, and nonstructural stormwater BMPs.

Appendix H. Maryland watersheds with Total Maximum Daily Loads for total suspended solids and similarities to Cabin John Creek.

Appendix I. Summary of urban stormwater BMP scenario and optimization studies.

Appendix J. Example WMOST v3.01 run set-ups

Calibration for TN (2006)

Calibration for TP (2006)

Optimization for TP (2003) including stormwater BMPs and riparian buffer

Optimization for TP (2014) including stormwater BMPs and riparian buffer